TU/e, 2MBA70

Solutions to problems for Measure and Probability Theory

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2 Measurable spaces (sigma-algebras and measures)

Problem 2.3

(a)
1. From the definition it is obvious that $\emptyset, \Omega \in \mathcal{F}$.
2. Since $\Omega \setminus \emptyset = \Omega$ and $\Omega \setminus \Omega = \emptyset$, the second property holds as well.
3. The only possible sequence is $\emptyset, \Omega$ whose union is $\Omega \in \mathcal{F}$.

(b)
1. From the definition it is obvious that $\emptyset, \Omega \in \mathcal{F}$.
2. The only additional thing to check with respect to (a) is to observe that $A^c = \Omega \setminus A \in \mathcal{F}$ and $(A^c)^c = A \in \mathcal{F}$.
3. The only other outcomes for sequences of sets with respect to (a) are A and A^c, which are both in $\mathcal{F}$.

(c) Since $\mathcal{P}$ contains all sets, it trivially satisfies all requirements.

(d) The key property that is violated is 2., since $A^c = \Omega \setminus A \notin \mathcal{F}$ since $\emptyset \neq A \subset \Omega$.

(e) Suppose that $\mathcal{F}$ is a σ-algebra and consider the countable family A_n, with $A_1 = \emptyset$ and $A_n = [0, 1/2 - 1/n] \cup [1/2 + 1/n, 1]$ for $n \geq 2$. Note that since $1/2 \notin A_n$ for all n it follow that $A_\infty := \bigcup_{n \in \mathbb{N}} A_n = [0, 1] \setminus \{1/2\}$. Since we assumed that $\mathcal{F}$ was a σ-algebra it must then hold that $\{1/2\} = [0, 1] \setminus A_\infty \in \mathcal{F}$. However, there is no way we can get a singleton with finite unions of the given intervals. Thus we arrive at a contradiction and conclude that $\mathcal{F}$ cannot be a σ-algebra.

(f)
1. Note that $f^{-1}(\emptyset) = \emptyset$ and $f^{-1}(\Omega') = \Omega$.
2. Here we use that
$$\begin{aligned} f^{-1}(\Omega' \setminus A') &= \{\omega \in \Omega \,:\, f(\omega) \notin A'\} \\ &= \Omega \setminus f^{-1}(A') \in \mathcal{F}. \end{aligned}$$
3. For any countable family A'_n we have that
$$\begin{aligned} f^{-1}(\bigcup_{n \in \mathbb{N}} A'_n) &= \{\omega \in \Omega \,:\, f(\omega) \in \bigcup_{n \in \mathbb{N}} A'_n\} \\ &= \bigcup_{n \in \mathbb{N}} \{\omega \in \Omega \,:\, f(\omega) \in A_n\} \\ &= \bigcup_{n \in \mathbb{N}} f^{-1}(A_n) \in \mathcal{F}. \end{aligned}$$

Problem 2.5
It suffices to show that $\sigma(\mathcal{A}) \subseteq \sigma(\mathcal{B})$. Assume now that $\sigma(\mathcal{B}) \subsetneq \sigma(\mathcal{A})$ and note that $\mathcal{A} \subset \mathcal{B}$ implies that $\mathcal{A} \subset \sigma(\mathcal{B})$. This then contradicts the fact that $\sigma(\mathcal{A})$ is the smallest σ-algebra that contains $\mathcal{A}$ which finishes the proof.

Problem 2.6
Let $\mathcal{O}$ denote the open sets in $\mathbb{R}^d$.

(a) Note that the set $(a_1, b_1) \times \cdots \times (a_d, b_d)$ is open for any $a_i < b_i \in \mathbb{R}$.

Hence $\mathcal{A}_1 \subset \mathcal{A}_1' \subset \mathcal{O}$ and thus by Lemma 2.5 we have that $\sigma(\mathcal{A}_1) \subset \sigma(\mathcal{A}_1') \subset \sigma(\mathcal{O}) = \mathcal{B}_{\mathbb{R}}$.

(b) The inclusion $\supset$ is trivial. So assume that $x \in O$. Then by definition there exist an $r > 0$ such that the ball $B_x(r) \subset O$. Now note that for any such d-dimensional ball, we can construct intervals (a_i, b_i) such that $I := (a_1, b_1) \times \cdots \times (a_d, b_d) \subset B_x(r)$ and $x \in I$. Thus $x \in \bigcup_{I \in \mathcal{A}, I \subset O} I$.

(c) Take $O \in \mathcal{O}$. If we can show that $O \in \sigma(\mathcal{A})$ then $\mathcal{B}_{\mathbb{R}} = \sigma(\mathcal{O}) \subset \sigma(\mathcal{A})$. The result then follows from 1.

From 2 it follows that O is a union over a subset collection of interval $(a_1, b_1) \times \cdots \times (a_d, b_d)$ where $a_i, b_i \in \mathbb{Q}$. Since $\mathbb{Q}$ is countable and d finite, the collection $\{(a_1, b_1) \times \cdots \times (a_d, b_d) : a_i < b_i \in \mathbb{Q}\}$ is also countable and hence $O = \bigcup_{I \in \mathcal{A}, I \subset O} I \in \sigma(\mathcal{A})$, from which it follows that $\mathcal{B}_{\mathbb{R}} \subset \sigma(\mathcal{A})$.

(d) This follows immediately from 1 and 3 since these imply that $\mathcal{B}_{\mathbb{R}} = \sigma(\mathcal{A}_1) \subset \sigma(\mathcal{A}_1') \subset \mathcal{B}_{\mathbb{R}}$.

(e) The inclusion $\subset$ is trivial, since $(a, b] \subset (a + b + 1/j)$ for any $a < b \in \mathbb{R}$ and $j \in \mathbb{N}$. For the other inclusion we argue by contradiction. Suppose that

$$x \in \bigcap_{j \in \mathbb{N}} (a_1, b_1 + \frac{1}{j}) \times \cdots \times (a_d, b_d + \frac{1}{j})$$

but $x \notin (a_1, b_1] \times \cdots \times (a_d, b_d]$. Then, writing $x = (x_1, \ldots, x_d)$, it must hold that $x_i > b_i$ for some $1 \leq i \leq d$ and hence there exists a $j \in \mathbb{N}$ such that $(b_i - x_i) > 1/j$. But this implies that $x_i \notin (a_i, b_i + 1/j)$ which is a contradiction. So we conclude that

$$(a_1, b_1] \times \cdots \times (a_d, b_d] \supset \bigcap_{j \in \mathbb{N}} (a_1, b_1 + \frac{1}{j}) \times \cdots \times (a_d, b_d + \frac{1}{j}).$$

(f) This time the inclusion $\supset$ is trivial since $(a, b - 1/j] \subset (a, b)$ holds for every $a < b \in \mathbb{R}$ and $j \in \mathbb{N}$. For the other inclusion suppose that $x \in (a_1, b_1) \times \cdots \times (a_d, b_d)$. Then, writing again $x = (x_1, \ldots, x_d)$, there exists a $j \in \mathbb{N}$ such that

$$x + 1/j := (x_1 + 1/j, \ldots .x_d + 1/j) \in (a_1, b_1) \times \cdots \times (a_d, b_d).$$

In particular, this implies that $b_i - (x_i + 1/j) > 0$ for every $1 \le i \le d$, which implies that $x \in (a_1, b_1 - 1/j] \times \cdots \times (a_d, b_d - 1/j]$ and hence

$$x \in \bigcup_{j \in \mathbb{N}} (a_1, b_1 - \frac{1}{j}] \times \cdots \times (a_d, b_d - \frac{1}{j}].$$

(g) It is clear that $\mathcal{A}_2 \subset \mathcal{A}_2'$. By 5 it follows that any interval $(a_1, b_1] \times \cdots \times (a_d, b_d]$ can be obtained as a countable intersection of intervals of the form $(a_1, b_1 + 1/j) \times \cdots \times (a_d, b_d + 1/j)$. By 4 $\mathcal{B}_{\mathbb{R}} = \sigma(\mathcal{A}_1')$ which by Lemma 2.1.2 contains

$$\bigcap_{j \in \mathbb{N}} (a_1, b_1 + 1/j) \times \cdots \times (a_d, b_d + 1/j) = (a_1, b_1] \times \cdots \times (a_d, b_d].$$

So we conclude that any interval $(a_1, b_1] \times \cdots \times (a_d, b_d] \in \mathcal{B}_{\mathbb{R}}$ from which it now follows that

$$\sigma(\mathcal{A}_2) \subset \sigma(\mathcal{A}_2') \subset \sigma(\mathcal{A}_1') = \mathcal{B}_{\mathbb{R}}.$$

For the other inclusion we consider a set $I := (a_1, b_1) \times \cdots \times (a_d, b_d)$ with $a_i < b_i \in \mathbb{Q}$. Then by 6 we have that $I = \bigcup_{j \in \mathbb{N}} (a_1, b_1 - 1/j] \times \cdots \times (a_d, b_d - 1/j]$ where the later is a countable union of sets $(a_1, c_1] \times \cdots \times [a_d, c_d]$ with $a_i < c_i \in \mathbb{Q}$ which must be in $\sigma(\mathcal{A}_2)$ by definition of a σ-algebra. Hence, any such interval $I \in \sigma(\mathcal{A}_2)$ and we thus conclude, using 3, that

$$\mathcal{B}_{\mathbb{R}} = \sigma(\mathcal{A}_1) \subset \sigma(\mathcal{A}_2) \subset \sigma(\mathcal{A}_2') \subset \sigma(\mathcal{A}_1') = \mathcal{B}_{\mathbb{R}},$$

which implies the result.

(h) (2 points) Step 1 is to show that any interval $[a_1, b_1) \times \cdots \times [a_d, b_d)$ can be obtained as a countable intersection of intervals $(a_1 - 1/j, b_1) \times \cdots \times (a_d - 1/j, b_d)$. From this we can conclude that any such must be in $\mathcal{B}_{\mathbb{R}}$ proving inclusions $\subset$.

For the other inclusions we have to show that any interval $(a_1, b_1) \times \cdots \times (a_d, b_d)$ can be obtained as a countable union of intervals $[a_1 + 1/j, b_1) \times \cdots \times [a_d + 1/j, b_d)$, which implies that such sets must be in the σ-algebra generated by $[a_1, b_1) \times \cdots \times [a_d, b_d)$.

(i) The main tool is to show that each of the intervals in these families can be obtained by taking a collection of allowed set operations for σ-algebras, i.e. countable unions/intersections and finite complements. This will help use prove the $\subset$ inclusions.

Then we show that any set from one of the first three families can also be obtained through countable unions/intersections and finite complements of sets from the last four families. These will then yield the $\supset$ inclusions and finish the proof.

Problem 2.7

(a) 1. By definition $(a, a] = \emptyset \in \mathcal{S}$.

2. Let $A, B \in \mathcal{S}$. Then there are three non-trivial cases to consider

   Suppose that $A = (a, \infty)$ and $B = (b, \infty)$ and assume without loss of generality that $a \leq b$. Then $A \cap B = (b, \infty) \in \mathcal{S}$.

   Suppose $A = (a, b]$ and $B = (c, d]$. We can assume without loss of generality that $a \leq c$. Then
$$A \cap B = \begin{cases} (c, b] & \text{if } c < b \leq d \\ (c, d] & \text{if } c \leq d \leq b \\ \emptyset & \text{otherwise,} \end{cases}$$
   which are all part of $\mathcal{S}$.

   Finally suppose that $A = (a.b]$ and $B = (c, \infty)$. Then
$$A \cap B = \begin{cases} (a, b] & \text{if } c \leq a \\ (c, b] & \text{if } a \leq c \leq b \\ \emptyset & \text{otherwise,} \end{cases}$$
   which are all in $\mathcal{S}$.

3. Here there are four non-trivial cases to consider for A and B.

   Suppose that $A = (a, \infty)$ and $B = (b, \infty)$ and assume without loss of generality that $a \leq b$. Then $A \setminus B = (a, b] \in \mathcal{S}$.

   Suppose $A = (a, b]$ and $B = (c, d]$. We can assume without loss of generality that $a \leq c$. Then
$$A \setminus B = \begin{cases} (a, c] & \text{if } c < b \leq d \\ (a, c] \cup (d, b] & \text{if } c \leq d < b \\ (a, b] & \text{otherwise,} \end{cases}$$
   and each is a finite union of elements in $\mathcal{S}$.

   Finally suppose that $A = (a.b]$ and $B = (c, \infty)$. Then
$$A \setminus B = \begin{cases} \emptyset & \text{if } c \leq a \\ (a, c] & \text{if } a \leq c \leq b \\ (a, b] & \text{otherwise,} \end{cases}$$
   which are all in $\mathcal{S}$. On the other hand
$$B \setminus A = \begin{cases} (c, a\ \cup (b, \infty) & \text{if } c \leq a \\ (b, \infty) & \text{if } a \leq c \leq b \\ (c, \infty) & \text{otherwise,} \end{cases}$$
   and each is a finite union of elements in $\mathcal{S}$.

(b) Let $\mathcal{A}_1 := \{(a,b] : a \leq b \in \mathbb{R}\}$ and $\mathcal{A}_1 := \{(a,\infty) : a \in \mathbb{R}\}$ so that $\mathcal{S} = \mathcal{A}_1 \cup \mathcal{A}_2 \cup \mathbb{R}$. Note that by Proposition 2.7 $\mathcal{B}_\mathbb{R} = \sigma(\mathcal{A}_1) = \sigma(\mathcal{A}_2)$. By the inclusion property we then have that $\mathcal{B}_\mathbb{R} \subset \sigma(\mathcal{S})$. Observer further that $\sigma(\mathcal{A}_1 \cup \mathcal{A}_2) = \sigma(\mathcal{S})$. Since $\mathcal{B}_\mathbb{R}$ is the smallest σ-algebra containing $\mathcal{A}_1$ and also the smallest one containing $\mathcal{A}_2$ it must hold that

$$\sigma(\mathcal{S}) = \sigma(\mathcal{A}_1 \cup \mathcal{A}_2) \subset \mathcal{B}_\mathbb{R},$$

which finishes the proof.

Problem 2.9

First note that if $\mu(A \cap B) = \infty$ then by property 2 we have that also $\mu(A)$, $\mu(B)$ and $\mu(A \cup B) = \infty$ and hence the result holds trivially. So assume now that $\mu(A \cap B) < \infty$. Since

$$A \cup B = (A \setminus (A \cap B)) \cup (B \setminus (A \cap B)) \cup (A \cap B),$$

it follows from property 1 that

$$\mu(A \cup B) = \mu(A \setminus (A \cap B))) + \mu(A \cap B) + \mu(B \setminus (A \cap B)).$$

Adding $\mu(A \cap B) < \infty$ to both side we get

$$\begin{aligned} \mu(A \cup B) + \mu(A \cap B) &= \mu(A \setminus (A \cap B))) + \mu(A \cap B) + \mu(B \setminus (A \cap B)) + \mu(A \cap B) \\ &= \mu(A) + \mu(B), \end{aligned}$$

where the last line follows from applying property 3 twice.

Problem 2.10

(a) Define the sets E_i as follows:

$$E_1 = A_1, \quad E_k = A_k \setminus A_{k-1} \text{ for } k \geq 2.$$

Clearly, $E_k \cap E_t = \emptyset$ for any $k \neq t$.

For the other properties, note that $A_i = \bigcup_{k=1}^{i} E_k$ holds for $i = 1$. Suppose now that it holds for a given i. Then

$$\begin{aligned} A_{i+1} &= A_i \cup (A_{i+1} \setminus A_i) \\ &= \left(\bigcup_{k=1}^{i} E_k \right) \cup (A_{i+1} \setminus A_i) \\ &= \left(\bigcup_{k=1}^{i} E_k \right) \cup E_{i+1} = \bigcup_{k=1}^{i+1} E_k, \end{aligned}$$

and thus $A_i = \bigcup_{k=1}^{i} E_k$ holds for all $i \geq 1$ by induction.

This then immediately implies that $\bigcup_{i \geq 1} A_i = \bigcup_{k \geq 1} E_k$.

(b) By a) and σ-additivity we have that

$$\mu(A_i) = \mu\left(\bigcup_{k=1}^{i} E_k\right) = \sum_{k=1}^{i} \mu(E_k),$$

and

$$\mu\left(\bigcup_{i\geq 1} A_i\right) = \mu\left(\bigcup_{k\geq 1} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k).$$

Thus,

$$\lim_{i\to\infty} \mu(A_i) = \sum_{k=1}^{\infty} \mu(E_k) = \mu\left(\bigcup_{i\geq 1} A_i\right),$$

as required.

(c) Consider the family $B_i = A_1 \setminus A_i$.

Then $(B_i)_{i\geq 1}$ is an increasing family of sets satisfying $\bigcup_{i\geq 1} B_i = A_1$ and thus, by part 1 of Proposition 2.12 we have

$$\lim_{i\to\infty} \mu(B_i) = \mu\left(\bigcup_{i\geq 1} B_i\right).$$

Since $\mu(A_1) < \infty$, the exclusion property (point 3 in Proposition 2.11) gives us that

$$\lim_{i\to\infty} \mu(B_i) = \mu(A_1) - \lim_{i\to\infty} \mu(A_i).$$

On the other hand

$$\bigcup_{i\geq 1} B_i = \bigcup_{i\geq 1} (A_1 \setminus A_i) = A_1 \setminus \bigcap_{i\geq 1} A_i$$

and thus

$$\mu\left(\bigcup_{i\geq 1} B_i\right) = \mu\left(A_1 \setminus \bigcap_{i\geq 1} A_i\right) = \mu(A_1) - \mu\left(\bigcap_{i\geq 1} A_i\right),$$

where we used the exclusion property since $\mu\left(\bigcap_{i\geq 1} A_i\right) \leq \mu(A_1) < \infty$.

Together we get

$$\mu(A_1) - \lim_{i\to\infty} \mu(A_i) = \mu(A_1) - \mu\left(\bigcap_{i\geq 1} A_i\right),$$

which yields the required identity since $\mu(A_1) < \infty$.

Problem 2.11
The idea is to construct a family of disjoint sets $(E_i)_{i\in\mathbb{N}}$ with the following properties:

1. $E_i \subset A_i$, and
2. $\bigcup_{i\in\mathbb{N}} E_i = \bigcup_{i\in\mathbb{N}} A_i$.

If such a sequence exists then we have

$$\begin{aligned}
\mu(\bigcup_{i\in\mathbb{N}} A_i) = \mu(\bigcup_{i\in\mathbb{N}} E_i) & \qquad \text{by 2} \\
= \sum_{i=1}^{\infty} \mu(A_i) & \qquad \text{because } E_i \text{ are disjoint and } \mu \text{ is } \sigma\text{-additive} \\
\leq \sum_{i=1}^{\infty} \mu(A_i) & \qquad \text{by 1 and monotone property of } \mu.
\end{aligned}$$

So we are left to construct the required family of sets $(E_i)_{i\in\mathbb{N}}$. The following set will do:

$$E_1 = A_1 \quad E_i = A_i \setminus \bigcup_{k<i}^{i} A_k \text{ for all } i > 1.$$

Note that by definition the set E_i are pair-wise disjoint and property1 holds. Finally, property 2 holds since $\bigcup_{i=1}^{k} E_i = \bigcup_{i=1}^{k} A_i$ holds for all $k \geq 1$.

Problem 2.13

(a) We first make the following observations about $\mathcal{N}$:

- because $\mu(\emptyset) = 0$ it holds that $\emptyset \in \mathcal{N}$,
- if $N, M \in \mathcal{N}$ then $N \setminus M \in \mathcal{N}$ since $N \setminus M \subset N$, and
- if $(N_i)_{i\geq 1}$ is a family of sets in $\mathcal{N}$ then so is $\bigcup_{i\geq 1} N_i$.

From the first point it follows that $\emptyset = \emptyset \cup \emptyset \in \overline{\mathcal{F}}$ and $\Omega = \Omega \cup \emptyset \in \overline{\mathcal{F}}$.

Furthermore, if $A, B \in \mathcal{F}$ and $N, M \in \mathcal{N}$, then by the second point and because $A \setminus B \in \mathcal{F}$,

$$(A \cup N) \setminus (B \cup M) = (A \setminus B) \cup (N \setminus M) \in \overline{\mathcal{F}}.$$

Finally, let $(A_i \cup N_i)_{i\geq 1}$ be a collection of sets in $\mathcal{N}$. Then using the third point we get

$$\bigcup_{i\geq 1} A_i \cup N_i = \bigcup_{i\geq 1} A_i \cup \bigcup_{i\geq 1} N_i \in \overline{\mathcal{F}}.$$

(b) (1 pt) From the definition we immediately get that $\mu(\emptyset) = 0$. Now, let $(A_i \cup N_i)_{i\geq 1}$ be a collection of disjoint sets in $\mathcal{N}$. Then

$$\bar{\mu}(\bigcup_{i\geq 1} A_i \cup N_i) = \bar{\mu}(\bigcup_{i\geq 1} A_i \cup \bigcup_{i\geq 1} N_i) = \mu(\bigcup_{i\geq 1} A_i) = \sum_{i\geq 1} \mu(A_i) = \sum_{i\geq 1} \bar{\mu}(A_i \cup N_i).$$

(c) This follows from the fact that $\bar{\mu}|_{\mathcal{F}}(A) = \bar{\mu}(A \cup \emptyset) = \mu(A)$.

(d) Suppose that $N \subset \Omega$ is a null set for $\overline{\mathcal{F}}$. Then there exists an $A \cup M \in \overline{\mathcal{F}}$ such that $N \subset A \cup M$ and $\bar{\mu}(A \cup M) = \mu(A) = 0$. However, since $M \in \mathcal{N}$, there must also exist a $B \in \mathcal{F}$ with $M \subset B$ and $\mu(B) = 0$. But this implies that $N \subset A \cup B \in \mathcal{F}$ which implies that $N \in \mathcal{N}$. Therefore, since $N = \emptyset \cup N$ it follows that $N \in \overline{\mathcal{F}}$ and hence every null set is part of the σ-algebra.

3 Measurable functions

Problem 3.2

(a) First we note that $f^{-1}(\emptyset) = \emptyset \in \mathcal{F}$ and $f^{-1}(E) = \Omega \in \mathcal{F}$. So $\emptyset, E \in \mathcal{H}$.

Next, let $B \in \mathcal{H}$. Then

$$f^{-1}(E \setminus B) = \Omega \setminus f^{-1}(B) \in \mathcal{F},$$

since by definition $f^{-1}(B) \in \mathcal{F}$. So $E \setminus B \in \mathcal{H}$.

Finally, if $(B_i)_{i\in\mathbb{N}}$ is a sequence of sets in $\mathcal{H}$, then

$$f^{-1}\left(\bigcup_{i=1}^{\infty} B_i\right) = \bigcup_{i=1}^{\infty} f^{-1}(B_i) \in \mathcal{F},$$

which shows that $\bigcup_{i=1}^{\infty} B_i \in \mathcal{H}$, completing the proof that $\mathcal{H}$ is a σ-algebra.

(b) By construction $\mathcal{A} \subseteq \mathcal{H}$. It therefore follows from Lemma 2.5 that $\mathcal{G} = \sigma(\mathcal{A}) \subseteq \mathcal{H}$. But this then implies that $f^{-1}(B) \in \mathcal{F}$ for each $B \in \mathcal{G}$ which means that f is $(\mathcal{F}, \mathcal{G})$-measurable.

Problem 3.3 It is clear that $f_{\#}\mu(\emptyset) = \mu(f^{-1}(\emptyset)) = \mu(\emptyset) = 0$. Suppose a sequence of mutually disjoint sets $B_i \in \mathcal{G}$, $i \in \mathbb{N}$, is given. Then,

$$f_{\#}\mu\left(\bigcup_{i=1}^{\infty} B_i\right) = \mu\left(f^{-1}\left(\bigcup_{i=1}^{\infty} B_i\right)\right) = \mu\left(\bigcup_{i=1}^{\infty} f^{-1}(B_i)\right) = \sum_{i=1}^{\infty} f_{\#}\mu(B_i).$$

Which shows that indeed $f_{\#}\mu$ is a measure on $\mathcal{G}$.
Now if μ is a probability measure on $\mathcal{F}$, it follows that $f_{\#}\mu(E) = \mu(f^{-1}(E)) = \mu(\Omega) = 1$, so $f_{\#}\mu$ is also a probability measure on $\mathcal{G}$.

Problem 3.5

(a) By Proposition 2.8, we know that $\mathcal{B}_{\mathbb{R}}$ is generated by different families of intervals. By Lemma 3.4 it thus suffices to check measurability only on one of these generating families.

As we saw in the proof of points (2) and (4) we should try to express a given preimage as a countable union of sets of which we already know are measurable.

(b) By Proposition 2.8, we know that $\mathcal{B}_\mathbb{R}$ is generated by intervals of the form $(-\infty, a]$ with $a \in \mathbb{Q}$. As a consequence, $\mathcal{B}_\mathbb{R}$ is also generated by intervals of the form $(a, +\infty)$ with $a \in \mathbb{Q}$. Therefore, by Lemma 3.1.4, it suffices to show that the set

$$\big\{\omega \in \Omega : f(\omega) + g(\omega) \in (a, +\infty)\big\}$$

is measurable for every $a \in \mathbb{Q}$. For brevity, we write $\{f + g > a\}$.

We will show that

$$\{f + g > a\} = \bigcup_{t \in \mathbb{Q}} \Big(\{f > t\} \cap \{g > a - t\}\Big),$$

which proves the claim as the sets on the right hand side are all measurable.

We first show the inclusion '$\subset$'. If $\omega \in \Omega$ is such that

$$f(\omega) + g(\omega) > a,$$

then

$$f(\omega) > a - g(\omega),$$

so there exists some $t \in \mathbb{Q}$ such that

$$f(\omega) > t > a - g(\omega),$$

and thus $f(\omega) > t$ and $g(\omega) > a - t$. So in that case

$$\omega \in \bigcup_{t \in \mathbb{Q}} \Big(\{f > t\} \cap \{g > a - t\}\Big).$$

Now we will show the inclusion '$\supset$'. Let $\omega \in \Omega$ be such that $f(\omega) > t$ and $g(\omega) > a - t$. Then, by adding the inequalities, we know that $f(\omega) + g(\omega) > a$.

(c) The constant function $f(\omega) = a$ is measurable since

$$f^{-1}(B) = f^{-1}(B \cap \{a\}) \cup f^{-1}(B \setminus \{a\}) = \Omega \cup \emptyset = \Omega \in \mathcal{F} \qquad \forall\, B \in \mathcal{B}_\mathbb{R}.$$

(d) Similar to the proof of Point (2) of Proposition 3.2.12.

(e) Let $g(\omega) \neq 0$ for all $\omega \in \Omega$. Then, since g is measurable, we have that

$$\begin{aligned}\{1/g > a\} &= \{g < 1/a,\ g > 0\} \cup \{g > 1/a,\ g < 0\} \\ &= \Big(\{g < 1/a\} \cap \{g > 0\}\Big) \cup \Big(\{g > 1/a\} \cap \{g < 0\}\Big) \in \mathcal{F},\end{aligned}$$

thus implying that $1/g$ is measurable.

(f) The previous part of this exercise together with point (4) of Proposition 3.12 yields Point (5) of Proposition 3.12.

Problem 3.6

(a) From (3.1), we have for any $a \in \mathbb{R}$,

$$\left\{\sup_{n\geq 1} f_n > a\right\} = \bigcup_{n\geq 1}\{f_n > a\},$$

Since f_n is measurable for all $n \geq 1$ it follows from Proposition 3.10 that $\{f_n > a\} \in \mathcal{F}$ for all $n \geq 1$.

Since $\mathcal{F}$ is a σ-algebra we then conclude that $\left\{f_n > a\right\} \in \mathcal{F}$,

which shows that

$$\left\{\sup_{n\geq 1} f_n > a\right\} \in \mathcal{F}.$$

Since the sets (a, ∞) generate $\mathcal{B}$ by Proposition 2.7 we conclude that $\sup_{n\geq 1} f_n$ is measurable.

(b) **(2)** Recall that $\inf_{n\geq 1} f_n(\omega) = -\sup_{n\geq 1}(-f_n(\omega))$.

If $f : \Omega \to \mathbb{R}$ is measurable then so is $-f$, since

$$(-f)^{-1}((t,\infty)) = \{\omega \in \Omega \,:\, t < -f(\omega) < \infty\} = \{\omega \in \Omega \,:\, -\infty < f(\omega) \leq t\} = f^{-1}((-\infty, t]) \in \mathcal{F}.$$

Thus, by part 1 of Lemma 3.12 we conclude that $\inf_{n\geq 1} f_n$ is measurable. **(3)** Let $\omega \in \Omega$ then

$$\limsup_{n\to\infty} f_n(\omega) := \inf_{n\geq 1}\sup_{m\geq n} f_m(\omega).$$

By part 1 of Lemma 3.12 the function $g_n(\omega) := \sup_{m\geq n} f_m(\omega)$ is measurable for any n.

Therefore, by part 2 of Lemma 3.12 $f_n = \inf_{n\geq 1} g_n$ is measurable and thus so is $\limsup_{n\to\infty} f_n$. **(4)** This follows directly from the previous part and the fact that $-\inf_{n\geq 1} f_n(\omega) = \sup_{n\geq 1}(-f_n(\omega))$.

(c) If the limit exists it must be equal to both the $\limsup$ and $\liminf$. Thus, measurability follows from point 3 or 4 of Lemma 3.12.

Problem 3.7

(a) Note that

$$f_M = M\mathbf{1}_{\{f\geq M\}} + f\mathbf{1}_{\{|f|<M\}} - M\mathbf{1}_{\{f\leq -M\}}.$$

Since the sets

$$\{f \geq M\}, \quad \{f \leq -M\}, \quad \{|f| < M\} \qquad \text{are } \mathcal{F}\text{-measurable},$$

their corresponding indicator functions are $(\mathcal{F}, \mathcal{B}_{\mathbb{R}})$-measurable.

Since f_M is the sum of products of $(\mathcal{F}, \mathcal{B}_{\mathbb{R}})$-measurable functions, we conclude that f_M is also $(\mathcal{F}, \mathcal{B}_{\mathbb{R}})$-measurable.

(b) We first show that f_M converges pointwise to f as $M \to \infty$, i.e.,

$$\lim_{M\to\infty} f_M(\omega) = f(\omega) \qquad \forall\, \omega \in \Omega.$$

Indeed, if $\omega \in \Omega$ is such that $f(\omega) = +\infty$, then

$$\lim_{M\to\infty} f_M(\omega) = \lim_{M\to\infty} M = +\infty = f(\omega),$$

and similarly for $\omega \in \Omega$ for which $f(\omega) = -\infty$.

On the other hand, for any $\omega \in \Omega$ with $f(\omega) \in \mathbb{R}$, there is some $N_0(\omega) \in \mathbb{N}$ such that $f_N(\omega) = f(\omega)$ for all $N \geq N_0(\omega)$, and hence,

$$\lim_{M\to\infty} f_M(\omega) = f(\omega).$$

Since f is the limit of a sequence of $(\mathcal{F}, \mathcal{B}_{\mathbb{R}})$-measurable functions, we conclude from Lemma 3.12 that f is $(\mathcal{F}, \mathcal{B}_{\mathbb{R}})$-measurable.

4 The Lebesgue Integral

Problem 4.2
We first make the following observation: Suppose that $A_i \cap B_j \neq \emptyset$ and take $\omega \in A_i \cap B_j$. Then $a_i = f(\omega) = b_j$.
We thus conclude that $a_i = b_j$ must hold for any i, j such that $A_i \cap B_j \neq \emptyset$.
Using that $A_i = \bigcup_{j=1}^M A_i \cap B_j$ and similarly $B_j = \bigcup_{i=1}^N A_i \cap B_j$, and these are all disjoint sets, we get

$$\begin{aligned}
\sum_{i=1}^N a_i \mu(A_i) &= \sum_{i=1}^N a_i \mu(\bigcup_{j=1}^M A_i \cap B_j) \\
&= \sum_{i=1}^N a_i \sum_{j=1}^M \mu(A_i \cap B_j) \\
&= \sum_{i=1}^N \sum_{j=1}^M a_i \mu(A_i \cap B_j) \\
&= \sum_{i=1}^N \sum_{j=1}^M b_i \mu(A_i \cap B_j) \\
&= \sum_{j=1}^M b_j \mu(\bigcup_{i=1}^N A_i \cap B_j) = \sum_{j=1}^M b_j \mu(B_j).
\end{aligned}$$

Problem 4.3

(a) The fact that the sets are disjoint is immediate from the definition. Measurability follows from Lemma 3.11

(b) Let us fix a $\omega \in \Omega$. Then if $f(\omega) = +\infty$ we get that $f_n(\omega) = 2^n$ holds for all $n \geq 1$ and hence $\lim_{n\to\infty} f_n(\omega) = +\infty = f(\omega)$. So assume that $f(\omega) < +\infty$. Then there exists an $M \in \mathbb{N}$ such that $f(\omega) < M$. Hence, for all $n \geq M$ we have that

$$\|f_n(\omega) - f(\omega)\| = f(\omega) - f_n(\omega) \leq 2^{-n},$$

which implies that $\lim_{n\to\infty} f_n(\omega) = f(\omega)$.

(c) Fix $n \geq 1$ and $\omega \in \Omega$. Clearly, if $f(\omega) = +\infty$ then $f_n(\omega) = 2^n < +\infty = f(\omega)$.

(d) Fix $\omega \in \Omega$ such that $f(\omega) < +\infty$ and $\omega \in A_k^n$ for some $0 \le k < N_n = n2^n$.

Note that $k2^{-n} \le f(\omega) < (k+1)2^{-n}$ holds and this interval can be split into two intervals as follows:

$$[k2^{-n}, (k+1)2^{-n}) = [(2k)2^{-(n+1)}, (2k+1)2^{-(n+1)}) \cup [(2k+1)2^{-(n+1)}, (2k+2)2^{-(n+1)}).$$

Hence, we conclude that either $\omega \in A_{2k}^{n+1}$ or $\omega \in A_{2k+1}^{n+1}$. In both case we get that

$$f_n(\omega) = k2^{-n} = 2kn^{-(n+1)} \le f_{n+1}(\omega).$$

(e) Now let us consider the case where $\omega \in A_k^n$ with $k = n2^n$, so that $n \le f(\omega) < +\infty$. Then, if $f(\omega) \ge n+1$ it follows that $f_n(\omega) = n < n+1 = f_{n+1}(\omega)$. If, on the other hand, $n \le f(\omega) < n+1$ there exists an $2n\,2^n \le \ell \le (2n+2)\,2^n$ such that $\omega \in A_\ell^{n+1}$, which then implies that

$$f_n(\omega) = n = (2n2^n)\,2^{-(n+1)} \le f_{n+1}(\omega).$$

Problem 4.5

1. First suppose $f = \sum_{i=1}^N a_i \mathbb{1}_{A_i}$ is a simple function. Then $f\mathbb{1}_B = \sum_{i=1}^N a_i \mathbb{1}_{A_i \cap B}$ is also a simple function and thus

$$\int_B f \,\mathrm{d}\mu = \int_\Omega f\mathbb{1}_B \,\mathrm{d}\mu = \sum_{i=1}^N a_i \mu(A_i \cap B) \le \mu(B) \sum_{i=1}^N a_i \mu(A_i) = 0.$$

Now let f be a non-negative function and $g \le f$ be a simple function. Then $g\mathbb{1}_B \le f\mathbb{1}_B$ and thus by Definition 4.7

$$\int_B f \,\mathrm{d}\mu = \int_\Omega f\mathbb{1}_B \,\mathrm{d}\mu \ge \int_\Omega g\mathbb{1}_B \,\mathrm{d}\mu = 0,$$

which implies the result.

2. Suppose $f \le g$ are non-negative functions and observe that if h is a simple function such that $h \le f$ then also $h \le g$. Therefore we get

$$\int_\Omega f \,\mathrm{d}\mu = \sup_{h \le f} \left\{ \int_\Omega h \,\mathrm{d}\mu \right\} \le \sup_{h \le g} \left\{ \int_\Omega h \,\mathrm{d}\mu \right\} = \int_\Omega g \,\mathrm{d}\mu.$$

3. Suppose that h is a simple function. Then αh is also simple and it immediately follows that $\int_\Omega (\alpha h) \,\mathrm{d}\mu = \alpha \int_\Omega h \,\mathrm{d}\mu$. Now let f be non-negative. Then $h \le f \iff \alpha h \le \alpha f$

and $h \leq \alpha f \iff \alpha^{-1} h \leq f$. Thus by Definition 4.7 we have

$$\begin{aligned}
\alpha \int_\Omega f \, \mathrm{d}\mu &= \alpha \sup_{h \leq f} \left\{ \int_\Omega h \, \mathrm{d}\mu \right\} \\
&= \sup_{h \leq f} \alpha \left\{ \int_\Omega h \, \mathrm{d}\mu \right\} \\
&= \sup_{h \leq f} \left\{ \int_\Omega (\alpha h) \, \mathrm{d}\mu \right\} \\
&= \sup_{\alpha^{-1} h \leq f} \left\{ \int_\Omega h) \, \mathrm{d}\mu \right\} \\
&= \sup_{h \leq \alpha f} \left\{ \int_\Omega (\alpha h) \, \mathrm{d}\mu \right\} = \int_\Omega (\alpha f) \, \mathrm{d}\mu.
\end{aligned}$$

Problem 4.7

(a) Note that $(f \mathbb{1}_B)^+ = f^+ \mathbb{1}_B$ and $(f \mathbb{1}_B)^- = f^- \mathbb{1}_B$.

Then we get, using Lemma 4.8,

$$\int_B f \, \mathrm{d}\mu = \int_\Omega (f \mathbb{1}_B)^+ \, \mathrm{d}\mu - \int_\Omega (f \mathbb{1}_B)^- \, \mathrm{d}\mu = \int_\Omega f^+ \mathbb{1}_B \, \mathrm{d}\mu - \int_\Omega f^- \mathbb{1}_B \, \mathrm{d}\mu = 0 + 0 = 0.$$

(b) Here we note that if $f \leq g$ then $f^+ \leq g^+$, while $f^- \geq g^-$.

Hence, using Lemma 4.8 again,

$$\int_\Omega f \, \mathrm{d}\mu = \int_\Omega f^+ \, \mathrm{d}\mu - \int_\Omega f^- \, \mathrm{d}\mu \leq \int_\Omega g^+ \, \mathrm{d}\mu - \int_\Omega g^- \, \mathrm{d}\mu = \int_\Omega g \, \mathrm{d}\mu.$$

(c) Assume first that $\alpha \geq 0$. Then it follows from Lemma 4.8 that

$$\alpha \int_\Omega f^\pm \, \mathrm{d}\mu = \int_\Omega (\alpha f)^\pm \, \mathrm{d}\mu,$$

which implies the result.

Now suppose that $\alpha < 0$ so that $\beta := -\alpha > 0$. Note that for any function f we have

$$(-f)^+ := \max\{-f, 0\} = \min\{f, 0\} = f^-$$

and similarly $(-f)^- = f^+$.

We then get that

$$-\int_\Omega f \mathrm{d}\mu = -\int_\Omega f^+ \, \mathrm{d}\mu + \int_\Omega (f)^- \, \mathrm{d}\mu = -\int_\Omega (-f)^- \, \mathrm{d}\mu + \int_\Omega (-f)^+ \, \mathrm{d}\mu = \int_\Omega (-f) \, \mathrm{d}\mu.$$

The result then follows since

$$\alpha \int_\Omega f \, \mathrm{d}\mu = -\beta \int_\Omega f \, \mathrm{d}\mu = -\int_\Omega (\beta f) \, \mathrm{d}\mu = \int_\Omega (-\beta f) \, \mathrm{d}\mu = \int_\Omega (\alpha f) \, \mathrm{d}\mu.$$

(d) This result follows immediately from Lemma 4.8 and the observation that $(f+g)^{\pm} = f^{\pm} + g^{\pm}$.

Problem 4.8

(a) By definition, we have that $\nu_f(\Omega) = \int_\Omega f \,\mathrm{d}\mu = 1$.

Now let $(A_n)_{n\in\mathbb{N}}$ be a family of mutually disjoint measurable sets. Then we have that the sequence

$$g_n := \sum_{i=1}^{n} f\,\mathbf{1}_{A_i} = f\,\mathbf{1}_{\bigcup_{i=1}^{n} A_i} \;\longrightarrow\; g := f\,\mathbf{1}_{\bigcup_{i\in\mathbb{N}} A_i} \qquad \text{pointwise monotonically.}$$

By MCT, we then have that

$$\begin{aligned}
\nu_f\left(\bigcup_{i\in\mathbb{N}} A_i\right) &= \int_{\bigcup_{i\in\mathbb{N}} A_i} f \,\mathrm{d}\mu \\
&= \int_\Omega g \,\mathrm{d}\mu \\
&= \lim_{n\to\infty} \int_\Omega g_n \,\mathrm{d}\mu \\
&= \lim_{n\to\infty} \int_{\bigcup_{i=1}^{n} A_i} f \,\mathrm{d}\mu \\
&= \sum_{i\in\mathbb{N}} \nu_f(A_i),
\end{aligned}$$

thus showing that ν_f is a probability measure on $(\Omega, \mathcal{F})$.

(b) Following the hint, we start by considering nonnegative simple functions g. Suppose $g = \sum_{i=1}^{n} a_i \mathbf{1}_{A_i}$ for $a_i \in \mathbb{R}$ and $A_i \in \mathcal{F}$ mutually disjoint. Then,

$$\int_\Omega g \, d\nu_f = \sum_{i=1}^{n} a_i\, \nu_f(A_i) = \sum_{i=1}^{n} a_i \int_{A_i} f \,\mathrm{d}\mu = \int_\Omega g f \,\mathrm{d}\mu.$$

Now let g be a nonnegative measurable function and $[g]_n$ be a sequence of nonnegative simple functions that converge pointwise monotonically to g. Then MCT yields

$$\int_\Omega g \,\mathrm{d}\nu_f = \lim_{n\to\infty} \int_\Omega [g]_n \,\mathrm{d}\nu_f = \lim_{n\to\infty} \int_\Omega [g]_n f \,\mathrm{d}\mu = \int_\Omega g f \,\mathrm{d}\mu,$$

where we used the fact that $[g]_n f$ converges pointwise monotonically to gf.

(c) Let g be measurable. Then $g = g^+ - g^-$, where $g^\pm$ are nonnegative measurable functions. Since f is nonnegative, we have that $(fg)^\pm = fg^\pm$. Due to (b), we deduce

$$\int_\Omega g^\pm \,\mathrm{d}\nu_f = \int_\Omega g^\pm f \,\mathrm{d}\mu = \int_\Omega (gf)^\pm \,\mathrm{d}\mu.$$

Hence, $g^\pm$ is ν_f-integrable if and only if $(gf)^\pm$ is μ-integrable. Consequently, g is ν_f-integrable if and only if gf is μ-integrable, since

$$\int_\Omega |g| \,\mathrm{d}\nu_f = \int_\Omega g^+ \,\mathrm{d}\nu_f + \int_\Omega g^- \,\mathrm{d}\nu_f = \int_\Omega g^+ f \,\mathrm{d}\mu + \int_\Omega g^- f \,\mathrm{d}\mu = \int_\Omega |gf| \,\mathrm{d}\mu.$$

Problem 4.9

($\Rightarrow$) Let f be μ-integrable. Then both $|f|\mathbf{1}_{\{|f|<n\}}$ and $|f|\mathbf{1}_{\{|f|\geq n\}}$ are integrable, due to the monotonicity of the integral. By linearity of the integral,

$$\int_\Omega |f|\mathbf{1}_{\{|f|\geq n\}} \,\mathrm{d}\mu = \int_\Omega |f| \,\mathrm{d}\mu - \int_\Omega |f|\mathbf{1}_{\{|f|<n\}} \,\mathrm{d}\mu.$$

Since the sequence $g_n := |f|\mathbf{1}_{\{|f|<n\}} \geq 0$ converges pointwise monotonically to $|f|$, we can apply MCT to obtain

$$\lim_{n\to\infty} \int_\Omega |f|\mathbf{1}_{\{|f|<n\}} \,dd\mu = \int_\Omega |f| \,\mathrm{d}\mu.$$

Hence,

$$\lim_{n\to\infty} \int_\Omega |f|\mathbf{1}_{\{|f|\geq n\}} \,\mathrm{d}\mu = \int_\Omega |f| \,\mathrm{d}\mu - \lim_{n\to\infty} \int_\Omega |f|\mathbf{1}_{\{|f|<n\}} \,\mathrm{d}\mu = 0.$$

($\Leftarrow$) By assumption, there is some $N \geq 1$ such that

$$\int_\Omega |f|\mathbf{1}_{\{|f|\geq N\}} \,\mathrm{d}\mu \leq 1.$$

By linearity of the integral,

$$\int_\Omega |f| \,\mathrm{d}\mu = \int_\Omega |f|\mathbf{1}_{\{|f|<N\}} \,\mathrm{d}\mu + \int_\Omega |f|\mathbf{1}_{\{|f|\geq N\}} \,\mathrm{d}\mu \leq N\mu\big(\{|f| < N\}\big) + 1.$$

Since μ is a finite measure, the right-hand side is finite, implying that f is μ-integrable.

Problem 4.10
Observe that $\Omega = \bigcup_{n\in\mathbb{N}}\{|f| > n\}$.
We then get that

$$\sum_{n=1}^{\infty} \int_{\{|f|>n\}} |f| \,\mathrm{d}\mu = \int_\Omega |f| \,\mathrm{d}\mu < \infty.$$

This implies that for some N and all $n \geq N$: $\int_{\{|f|>n\}} |f| \,\mathrm{d}\mu < 1/n$ or else the sum cannot be finite.
Now let $\varepsilon > 0$, take $M > \max\{N, 2/\varepsilon\}$ and $\delta = \varepsilon/(2M)$. Then

$$\begin{aligned}\int_A |f| \,\mathrm{d}\mu &= \int_A |f| \mathbf{1}_{|f| \leq M} \,\mathrm{d}\mu + \int_A |f| \mathbf{1}_{|f|>M} \,\mathrm{d}\mu \\ &\leq M\mu(A) + \frac{1}{M} \leq M\delta + \frac{1}{M} < \varepsilon.\end{aligned}$$

5 Little Wood's Principles

Problem 5.1 (5 pts)
Let $\varepsilon > 0$ be arbitrary and $f \in L^1(\mathbb{R}^d, \mu)$. By Theorem 7.12, we find a bounded and continuous function $h \in L^1(\mathbb{R}^d, \mu)$ such that $\|f - h\|_1 \leq \varepsilon/3$. (0.5 pts)
Let $K > 0$ and consider the function

$$h_K := h\mathbf{1}_{E_K} \in L^1(\mathbb{R}^d, \mu), \qquad E_K := [-K, K]^d,$$

i.e., h_K is a horizontal truncation of h to the compact set $E_K \subset \mathbb{R}^d$ (1 pts).
Then,

$$\int_{\mathbb{R}^d} |h - h_K| \, d\mu = \int_{E_K^c} |h| \, d\mu \leq \|h\|_\infty \, \mu(E_K^c). \qquad \text{(1 pts)}$$

Since μ is a finite measure, we have that $\lim_{K\to\infty} \mu(E_K^c) = 0$. In particular, we can make the rhs of the estimate above arbitrarily small such that $\|h\|_\infty \, \mu(E_K^c) \leq \varepsilon/3$. (0.5 pts)
We then find, by Theorem 7.12, a bounded and continuous function $g \in L^1(\mathbb{R}^d, \mu)$ such that $\|h_K - g\|_1 \leq \varepsilon/3$ (0.5 pts).
Since h_K is continuous on E_K, the function g may be taken to be h_K on E_K. Moreover, g can be chosen to have compact support since h_K has compact support. (0.5 pts)
Altogether, we obtain (1 pts)

$$\|f - g\|_1 \leq \|f - h\|_1 + \|h - h_K\|_1 + \|h_K - g\|_1 \leq \varepsilon/3 + \varepsilon/3 + \varepsilon/3 = \varepsilon.$$

6 Probability I, The basics

Problem 6.1

(a) Note that by assumption we have that

$$X_{\#}\mathbb{P}((-\infty,t]) := F_X(t) = F_Y(t) := Y_{\#}\mathbb{P}((-\infty,t]),$$

holds for all intervals $(-\infty,t]$. Moreover, these intervals generate the Borel σ-algebra, see Proposition 2.7. Therefore the idea is to use Theorem 2.15 to conclude that $X_{\#}\mathbb{P} = Y_{\#}\mathbb{P}$.

For this we need to check two properties for the generator set $\mathcal{A} = \{(-\infty,t] \,:\, t \in \mathbb{R}\}$.

1. Suppose $A = (-\infty,t]$ and $B = (-\infty,s]$ with $t \leq s$. Then $A \cap B = (-\infty,t] = A \in \mathcal{A}$.
2. Take $A_n = (\infty,n]$. Then $\bigcup_{n\in\mathbb{N}} A_n = \mathbb{R}$.

We thus conclude that $\mathcal{A}$ satisfies the requirements of Theorem 2.15 and thus $X_{\#}\mathbb{P} = Y_{\#}\mathbb{P}$ on the entire Borel σ-algebra.

(b) First note that there must exist a measurable set $N \in \mathcal{B}_{\mathbb{R}}$ such that $X_{\#}\mathbb{P}(N) = 0$. If not, then using the sequence of disjoint sets $A_n = [n, n+1)$ we get that

$$X_{\#}\mathbb{P}(\mathbb{R}) \geq X_{\#}\mathbb{P}(\bigcup_{n\in\mathbb{N}} A_n) = \sum_{n\in\mathbb{N}} X_{\#}\mathbb{P}(A_n) = \infty,$$

contradicting $X_{\#}\mathbb{P}(\mathbb{R}) = 1$.

Define the function $Y : \Omega \to \mathbb{R}$ as follows:

$$Y(\omega) = \begin{cases} X(\omega) & \text{if } \omega \in \mathbb{R} \setminus N, \\ X(\omega) + 2 & \text{if } \omega \in N. \end{cases}$$

Then clearly $Y \neq X$ as functions. However, for any measurable set A, writing $A\Delta N := \Omega \setminus (A \cap N)$, we have

$$\begin{aligned} X_{\#}\mathbb{P}(A) &= \mathbb{P}(X^{-1}(A)) \\ &= \mathbb{P}(X^{-1}(A\cap N) \cup X^{-1}(A\Delta N)) \\ &= \mathbb{P}(X^{-1}(A\cap N)) + \mathbb{P}(X^{-1}(A\Delta N)) \\ &= 0 + \mathbb{P}(\{\omega \in \Omega \,:\, X(\omega) \in A\Delta N\}) \\ &= 0 + \mathbb{P}(\{\omega \in \Omega \,:\, Y(\omega) \in A\Delta N\}) \\ &= \mathbb{P}(Y^{-1}(A\cap N)) + \mathbb{P}(Y^{-1}(A\Delta N)) \\ &= \mathbb{P}(Y^{-1}(A)) = Y_{\#}\mathbb{P}(A), \end{aligned}$$

so $X \overset{d}{=} Y$ still holds.

Problem 6.2

(a) The function $U(t)$ is measurable as it is a product of an indicator and a continuous function.

(b) By definition of U it follows that

$$U^{-1}((-\infty,t]) = \begin{cases} \emptyset & \text{if } t \leq 0, \\ (0,t] & \text{if } 0 < t \leq 1, \\ [0,1] & \text{if } t > 1. \end{cases}$$

(c) Since by Theorem 2.19

$$\lambda|_{[0,1]}((0,t]) = \lambda((0,t]) = t,$$

for any $0 < t \leq 1$ we have

$$\mathbb{P}\left(U^{-1}((-\infty,t])\right) := \lambda|_{[0,1]}\left(U^{-1}((-\infty,t])\right) = \begin{cases} 0 & \text{if } t < 0, \\ t & \text{if } 0 \leq t \leq 1, \\ 1 & \text{if } t > 1, \end{cases}$$

which proves the result.

Problem 6.3

(a) The implication from right to left is by definition of $\overleftarrow{F}$ and the fact that F is non-decreasing. The implication from left to right is because F is right continuous.

(b) Consider the preimage of $(-\infty,t]$ under X. Then, using the above observation, we have

$$\begin{aligned} X^{-1}((-\infty,t]) &= \{\omega \in \Omega \, : \, \overleftarrow{F}(U(\omega)) \in (-\infty,t]\} \\ &= \{\omega \in \Omega \, : , U(\omega) \in (-\infty, F(t)]\} = U^{-1}((-\infty,F(t)]) \in \mathcal{B}_{[0,1]}. \end{aligned}$$

Hence, X is measurable. Finally, the above computation, together with Lemma 6.5, also implies that

$$\mathbb{P}\left(X^{-1}((-\infty,t])\right) = \mathbb{P}\left(U^{-1}((-\infty,F(t)])\right) = F(t).$$

Now let $(\Omega,\mathcal{F},\mathbb{P})$ be a probability space and U a standard normal random variable. We will show that $X = \overleftarrow{F} \circ U$ is a random variable with the right probability measure. Since we can construct a standard uniform random variable on the probability $([0,1],\mathcal{B}_{[0,1]},\lambda|_{[0,1]})$ this also implies the last part.
which finished the proof.

Problem 6.4

(a) For the probability space, take $\Omega = [0, 1]$, $\mathcal{F} = \mathcal{B}_{[0,1]}$ and $\mathbb{P} = \lambda$ the Lebesgue measure restricted to $[0, 1]$.

Observe that the function $H_\gamma(z)$ is continuous and hence has an inverse $g_\gamma(y) = \gamma \tan(\pi(y - 1/2))$ on $[0, 1]$.

So the function $Y[0, 1] \to \mathbb{R}$ defined by $Y(x) = g_\gamma(x)$ has the correct distribution as

$$\mathbb{P}(Y^{-1}((-\infty, t])) = \mathbb{P}(g_\gamma^{-1}((-\infty, t])) = \lambda(H_\gamma((-\infty, t])) = H_\gamma(t).$$

(b) Note that g_γ is continuous on $[0, 1]$ and hence measurable.

(c) For any $t \geq 0$, the cdf of the Poisson random variable is given by

$$F_\lambda(t) = \sum_{n=0}^{\lceil t \rceil} f_\lambda(n),$$

where $\lceil t \rceil$ is the ceiling of t, i.e. the smallest integer $k \geq t$.

(d) For the probability space, we again take $\Omega = [0, 1]$, $\mathcal{F} = \mathcal{B}_{[0,1]}$ and $\mathbb{P} = \lambda$ the Lebesgue measure restricted to $[0, 1]$.

Now for any $y \in [0, 1]$ let $k := k(y)$ be such that

$$\sum_{n=1}^{k} f_\lambda(n) \geq y \quad \text{and} \quad \sum_{n=1}^{k-1} f_\lambda(n) < y,$$

where the last sum is interpreted as -1 if $k = 0$.

Now define $X(y) = k(y) : [0, 1] \to \mathbb{R}$. Then $k(y) \leq t$ if and only if $y \leq F_\lambda(t)$ and hence

$$X^{-1}((-\infty, t]) = \{y \in [0, 1] \,:\, k(y) \in (0, t]\} = \{y \in [0, 1] \,:\, y \in (0, F_\lambda(t)]\},$$

from which it follows that

$$\mathbb{P}(X^{-1}((-\infty, t])) = \lambda((0, F_\lambda(t)]) = F_\lambda(t).$$

(e) It follows from the above computation that $X^{-1}((-\infty, t]) = \{y \in [0, 1] \,:\, y \in (0, F_\lambda(t)]\}$. Since the latter is a measurable set we conclude that $X^{-1}((-\infty, t])$ is measurable for all t and since these generate the Borel σ-algebra X is measurable.

(f) For any $\ell \in \mathbb{N}$ define the sets $A_\ell = (n - 1 - 1/\ell, n - 1 + 1/\ell]$. Then A_ℓ is a decreasing set with $\lim_{\ell \to \infty} A_\ell = \{n - 1\}$. Moreover, $A_\ell = (-\infty, n - 1 + 1/\ell] \setminus (-\infty, n - 1 - 1/\ell]$ and $\mathbb{P}(A_1) < \infty$.

It now follows from continuity from above and (d) that

$$\begin{aligned}
X_\#\mathbb{P}(\{n\}) &= \lim_{\ell \to \infty} X_\#\mathbb{P}(A_\ell) \\
&= \lim_{\ell \to \infty} X_\#\mathbb{P}((-\infty, n - 1 + 1/\ell]) - X_\#\mathbb{P}((-\infty, n - 1 - 1/\ell]) \\
&= F_\lambda(n - 1 + 1/\ell) - F_\lambda(n - 1 - 1/\ell) \\
&= \sum_{k=0}^{n} f_\lambda(k) - \sum_{k=0}^{n-1} f_\lambda(k) = f_\lambda(n).
\end{aligned}$$

Problem 6.6

Define for any $j \in \mathbb{Z}$, $p_j := \mathbb{P}(X^{-1}(\{j\}))$. Then, since $(X^{-1}(j))_{j\in\mathbb{Z}}$ is a family of disjoint sets and $\mathbb{P}$ is a probability measure we get that

$$1 = \mathbb{P}(\Omega) = \mathbb{P}(\bigcup_{j\in\mathbb{Z}} X^{-1}(j)) = \sum_{j\in\mathbb{Z}} p_j.$$

Now let $A \subset \mathbb{R}$ be a measurable set and note that

$$X^{-1}(A) = \bigcup_{j\in\mathbb{Z}\cap A} X^{-1}(j).$$

Then it follows that

$$\mathbb{P}(X \in A) = \mathbb{P}(X^{-1}(A)) = \mathbb{P}\left(\bigcup_{j\in\mathbb{Z}\cap A} X^{-1}(j)\right) = \sum_{j\in\mathbb{Z}\cap A} p_j = \sum_{j\in\mathbb{Z}} \delta_j(A) p_j.$$

Problem 6.8

(a) We first observe that

$$\nu((-\infty, t]) = X_{\#}\mathbb{P}((-\infty, t])$$

holds by definition of the pdf.

Next we note that the collection $\mathcal{A} := \{(-\infty, t] \,:\, t \in \mathbb{R}\}$ generates $\mathcal{B}$ and satisfies the conditions of Theorem 2.15:

a) $(-\infty, t] \cap (-\infty, s] = (-\infty, \min\{t, s\}] \in \mathcal{A}$, and

b) the sets $A_n(-\infty, n] \in \mathcal{A}$ for $n \in \mathbb{N}$ satisfies $\bigcup_{n\in\mathbb{N}} A_n = \mathbb{R}$.

Thus, since $\nu = X_{\#}\mathbb{P}$ on $\mathcal{A}$, by Theorem 2.15 we conclude that $\nu = X_{\#}\mathbb{P}$ on $\sigma(\mathcal{A}_2) = \mathcal{B}$.

(b) Let $g = \sum_{i=1}^N a_i \mathbb{1}_{A_i}$ be a simple function. Then by definition of ν and linearity of the integral we get

$$\begin{aligned}
\int_{\mathbb{R}} g \,\mathrm{d}\nu &= \sum_{i=1}^N a_i \int_{A_i} \mathrm{d}\nu \\
&= \sum_{i=1}^N a_i \nu(A_i) \\
&= \sum_{i=1}^N a_i \int_{A_i} \rho \,\mathrm{d}\lambda \\
&= \int_{\Omega} \sum_{i=1}^N a_i \mathbb{1}_{A_i} \rho \,\mathrm{d}\lambda = \int_{\mathbb{R}} g\rho \,\mathrm{d}\lambda.
\end{aligned}$$

(c) By using (b) and monotone convergence twice we get

$$\begin{aligned}\int_{\mathbb{R}} h \, \mathrm{d}\nu &= \int_{\mathbb{R}} \lim_{n\to\infty} [h]_n \, \mathrm{d}\nu \\ &= \lim_{n\to\infty} \int_{\mathbb{R}} [h]_n \, \mathrm{d}\nu \\ &= \lim_{n\to\infty} \int_{\mathbb{R}} [h]_n \rho \, \mathrm{d}\lambda \\ &= \int_{\mathbb{R}} \lim_{n\to\infty} [h]_n \rho \, \mathrm{d}\lambda = \int_{\mathbb{R}} h\rho \, \mathrm{d}\lambda.\end{aligned}$$

(d) Using the change of variables result (Proposition 4.14) (a) and (c) we get

$$\mathbb{E}[h(X)] = \int_{\Omega} h \circ X \, \mathrm{d}\mathbb{P} = \int_{\mathbb{R}} h \, \mathrm{d}X_{\#}\mathbb{P} = \int_{\mathbb{R}} h \, \mathrm{d}\nu = \int_{\mathbb{R}} h\rho \, \mathrm{d}\lambda.$$

Problem 6.9

(a) This follows from the following computation

$$\int_{\Omega} |f|^p \, \mathrm{d}\mu \geq \int_{\Omega} |f|^p \mathbb{1}_{|f|\geq t} \, \mathrm{d}\mu \geq t^p \int_{\Omega} \mathbb{1}_{|f|\geq t} \, \mathrm{d}\mu = t^p \mu(\{\omega \in \Omega \,:\, |f| \geq t\}).$$

(b) Using the result for $p = 1$ we get

$$\mathbb{P}(|X| \geq t) = \mu(\omega \in \Omega \,:\, |X(\omega) \geq t\}) \leq \frac{1}{t} \int_{\Omega} X \, \mathrm{d}\mathbb{P} = \frac{1}{t}\mathbb{E}[X].$$

(c) Take $f(\omega) = X(\omega) - \mathbb{E}[X]$, which is measurable. Then using the first result with $p = 2$ gives

$$\begin{aligned}\mathbb{P}(|X - \mathbb{E}[X]| \geq t) &= \mathbb{P}(|X - \mathbb{E}[X]|^2 \geq t^2) \\ &\leq \frac{1}{t^2}\mathbb{E}[(X - \mathbb{E}[X])^2] \\ &= \frac{1}{t^2}(\mathbb{E}[X^2] - \mathbb{E}[X]^2) = \frac{\mathrm{Var}(X)}{t^2}.\end{aligned}$$

7 Product spaces and independent random variables

Problem 7.2 "$\subset$" By definition, the product σ-algebra $\mathcal{F}_1 \otimes \mathcal{F}_2$ is defined as the σ-algebra generated by the collection

$$\mathcal{A} := \Big\{ A \times B \subset \Omega_1 \times \Omega_2 : A \in \mathcal{F}_1,\ B \in \mathcal{F}_2 \Big\}.$$

Since $A \times B = (A \times \Omega_2) \cap (\Omega_1 \times B)$, we have that

$$A \times B = \pi_1^{-1}(A) \cap \pi_2^{-1}(B) \in \sigma(\pi_1, \pi_2).$$

"$\supset$" Let $C \in \{\pi_i^{-1}(A) : i = 1, 2,\ A \in \mathcal{F}_1\}$. Then there exist sets $A \in \mathcal{F}_1$ or $B \in \mathcal{F}_2$ such that $C = \pi_1^{-1}(A) = A \times \Omega_2$ or $C = \pi_2^{-1}(B) = \Omega_1 \times B$. Either way, since $\Omega_1 \in \mathcal{F}_1$ and $\Omega_2 \in \mathcal{F}_2$, we have that $C \in \mathcal{A}$.

Problem 7.3

(a) Note that $\mathcal{A}_1 \times \mathcal{A}_2 \subset \mathcal{F}_1 \times \mathcal{F}_2$, and hence

$$\sigma(\mathcal{A}_1 \times \mathcal{A}_2) \subset \mathcal{F}_1 \otimes \mathcal{F}_2.$$

(b) Let $B \in \mathcal{A}_2$. Then we have that

$$\Omega_1 \times B = \bigcup_{n \geq 1} A_n \times B \in \sigma(\mathcal{A}_1 \times \mathcal{A}_2)$$

since $A_n \times B \in \sigma(\mathcal{A}_1 \times \mathcal{A}_2)$ for all $n \geq 1$. So $\Omega_1 \in \Sigma$

For the second property, let $C \in \Sigma$ and note that $C^c \times B = (\Omega_1 \times B) \setminus (C \times B)$. Since both these sets are in $\sigma(\mathcal{A}_1 \times \mathcal{A}_2)$ it follows that $C^c \times B \in \sigma(\mathcal{A}_1 \times \mathcal{A}_2)$ and hence $C^c \in \Sigma$.

Finally consider a countable sequence $(C_n)_{n \geq 1}$ of sets in Σ. Then for any $B \in \mathcal{A}_2$

$$\left(\bigcup_{n \geq 1} C_n \right) \times B = \bigcup_{n \geq 1} (C_n \times B) \in \sigma(\mathcal{A}_1 \times \mathcal{A}_2),$$

since each $C_n \times B \in \sigma(\mathcal{A}_1 \times \mathcal{A}_2)$.

(c) Note that $\mathcal{A}_1 \subset \Sigma_1 \subset \mathcal{F}_1$. From which it follows that $\Sigma_1 = \mathcal{F}_1$.

But then, from the definition of Σ_1 we have that $\mathcal{F}_1 \times \mathcal{A}_2 \subset \sigma(\mathcal{A}_1 \times \mathcal{A}_2)$.

(d) We can show in a similar fashion that

$$\Sigma_2 := \{C \in \mathcal{F}_2 \,:\, B \times C \in \sigma(\mathcal{A}_1 \times \mathcal{A}_2)\, \forall B \in \mathcal{A}_1\}.$$

is a σ-algebra on Ω_2, from which we conclude that $\mathcal{A}_1 \times \mathcal{F}_2 \subset \sigma(\mathcal{A}_1 \times \mathcal{A}_2)$.

(e) Take any $A \in \mathcal{F}_1$ and $B \in \mathcal{F}_2$. Then

$$A \times B = (A \times \Omega_2) \cap (\Omega_1 \times B) = \bigcup_{n,m \geq 1} (A \times B_m) \cap (A_n \times B) \in \sigma(\mathcal{A}_1 \times \mathcal{A}_2).$$

From this we conclude that $\mathcal{F}_1 \times \mathcal{F}_2 \subset \sigma(\mathcal{A}_1 \times \mathcal{A}_2)$, which finishes the proof.

Problem 7.6
Assume that for all $a, b \in \mathbb{R}$

$$\mathbb{P}(X_1 \leq a, X_2 \leq b) = \mathbb{P}(X_1 \leq a)\mathbb{P}(X_2 \leq b).$$

We now have to show that X_1 and X_2 are independent according to Definition 7.7.
First note that since the family $(-\infty, a] \times (-\infty, b]$ generate the 2-dimensional Borel σ-algebra we have, using Theorem 2.2.17, that

$$\mathbb{P}(X_1 \in B_1, X_2 \in B_2) = \mathbb{P}(X_1 \in B_1)\mathbb{P}(X_2 \in B_2)$$

for all $B_1, B_2 \in \mathcal{B}_{\mathbb{R}}$.
Now fix a set $B_2 \in \mathcal{B}_{\mathbb{R}}$, set $A_2 := X_2^{-1}(B_2) \in \sigma(X_2)$, and define the following two measures on the space $(\Omega, \sigma(X_1))$

$$\mu_1(A) = \mathbb{P}(A \cap A_2) \quad \text{and} \quad \mu_2(A) = \mathbb{P}(A)\mathbb{P}(A_2).$$

Let $a \in \mathbb{R}$ and consider the set $A_1 := X_1^{-1}((-\infty, a]) \in \sigma(X_1)$. Then, by our assumption we have that

$$\mu_1(A_1) = \mathbb{P}(A_1 \cap A_2) = \mathbb{P}(X_1 \leq a, X_2 \in B_2) = \mathbb{P}(X_1 \leq a)\mathbb{P}(X_2 \in B_2) = \mathbb{P}(A_1)\mathbb{P}(A_2) = \mu_2(A_1).$$

In other words, the measures μ_1, μ_2 coincide on the set $\{X_1^{-1}((-\infty, a]) \,:\, a \in \mathbb{R}\}$.
Since the set $(-\infty, a]$ generate $\mathcal{B}_{\mathbb{R}}$ it follows that

$$\sigma(X_1) = \sigma(\{X_1^{-1}((-\infty, a]) \,:\, a \in \mathbb{R}\}).$$

In addition, this set satisfies the conditions of Theorem 2.2.17 and hence we conclude that $\mu_1(A) = \mu_2(A)$ for all $A \in \sigma(X_1)$.
We can repeat this argument for the two measures on $(\Omega, \sigma(X_2))$

$$\nu_1(A) = \mathbb{P}(A_1 \cap A) \quad \text{and} \quad \nu_2(A) = \mathbb{P}(A_1)\mathbb{P}(A),$$

where $A_1 \in \{X_1^{-1}((-\infty, a]) \,:\, a \in \mathbb{R}\}$ is fixed.
From this we conclude that for any $A_1 \in \sigma(X_1)$ and $A_2 \in \sigma(X_2)$

$$\mathbb{P}(A_1 \cap A_2) = \mathbb{P}(A_1)\mathbb{P}(A_2)$$

and hence X_1 and X_2 are independent.

8 Convergence of integrals and measures

Problem 8.1

Similar to the proof of Fatou's lemma, we define $g_n = \sup_{k\geq n} f_n$ which are measurable due to Proposition 3.13. Moreover, we have that $\limsup_{n\to\infty} f_n = \lim_{n\to\infty} g_n$.
Next we note that $g_n \geq f_\ell$ for all $\ell \geq n$. Thus, by monotonicity of the integral, we have that

$$\int_\Omega g_n \,\mathrm{d}\mu \geq \int_\Omega f_\ell \,\mathrm{d}\mu,$$

holds for all $\ell \geq n$, which implies that

$$\int_\Omega g_n \,\mathrm{d}\mu \geq \sup_{k\geq n} \int_\Omega f_n \,\mathrm{d}\mu.$$

In addition, since $g_n < f$ with f being non-negative and integrable we can apply Dominated Convergence to conclude that

$$\int_\Omega \lim_{n\to\infty} g_n \,\mathrm{d}\mu = \lim_{n\to\infty} \int_\Omega g_n \,\mathrm{d}\mu.$$

Putting all this together we get

$$\int_\Omega \limsup_{n\to\infty} f_n \,\mathrm{d}\mu = \int_\Omega \lim_{n\to\infty} g_n \,\mathrm{d}\mu = \lim_{n\to\infty} \int_\Omega g_n \,\mathrm{d}\mu \geq \lim_{n\to\infty} \sup_{k\geq n} \int_\Omega f_n \,\mathrm{d}\mu.$$

Problem 8.2

(a) Let $t_0 \in (a,b)$ be fixed. It suffices to check the continuity result for arbitrary sequences $(t_n)_{n\geq 1} \subset (a,b)$ such that $t_n \to t_0$ as $n \to \infty$. Fix such a sequence and define $g_n(\omega) := f(\omega, t_n)$ for all $\omega \in \Omega$ and $n \geq 1$. Since $\lim_{t\to t_0} f(\omega,t) = f(\omega,t_0)$ for all $\omega \in \Omega$, we deduce that $\lim_{n\to\infty} g_n(\omega) = f(\omega,t_0)$ for every $\omega \in \Omega$. Moreover, by assumption $|g_n| \leq g$ for all $n \geq 1$ and g is integrable. By the Dominated Convergence Theorem

$$\lim_{n\to\infty} \int_\Omega g_n(\omega)\,\mu(\mathrm{d}\omega) = \int_\Omega f(\omega,t_0)\,\mu(\mathrm{d}\omega).$$

As the chosen sequence was arbitrary, we deduce that $\lim_{t\to t_0} F(t) = F(t_0)$.

(b) If $t \mapsto f(\omega,t)$ is continuous on (a,b) for all $\omega \in \Omega$ then $\lim_{t\to t_0} f(\omega,t) = f(\omega,t_0)$ at every $t_0 \in (a,b)$ for all $\omega \in \Omega$. In particular, (a) applies, showing that $\lim_{t\to t_0} F(t) = F(t_0)$ for every $t_0 \in (a,b)$, i.e., F is continuous on (a,b).

Problem 8.3

(a) We start by showing that $(\partial f/\partial t)(\cdot, t)$ is measurable.

Let $(t_n)_{n\geq 1} \subset (a,b)$ be an arbitrary sequence with $t_n \neq t$ and $t_n \to t$ for $n \to \infty$. We set

$$h_n(\omega) = \frac{f(\omega, t_n) - f(\omega, t)}{t_n - t}.$$

Clearly, h_n is measurable for every $n \geq 1$.

Moreover, $\lim_{n\to\infty} h_n(\omega) = (\partial f/\partial t)(\omega, t)$ by the definition of the derivative. Since $(\partial f/\partial t)(\cdot, t)$ is the pointwise limit of a sequence of measurable functions, it is also measurable.

Finally, $(\partial f/\partial t)(\cdot, t)$ is integrable since

$$\int_\Omega |(\partial f/\partial t)(\omega, t)| \,\mu(\mathrm{d}\omega) \leq \int_\Omega g \,\mathrm{d}\mu < +\infty.$$

(b) Let $t_0 \in (a,b)$ and suppose w.l.o.g. $t_0 < t$. Since $t \mapsto f(\omega, t)$ is differentiable on (a,b) for all $\omega \in \Omega$, the Mean Value Theorem gives

$$\frac{f(\omega, t) - f(\omega, t_0)}{t - t_0} = (\partial f/\partial t)(\omega, \tau) \qquad \text{for some } \tau \in (t_0, t).$$

Taking the modulus on both sides, we obtain

$$\left|\frac{f(\omega, t) - f(\omega, t_0)}{t - t_0}\right| \leq |(\partial f/\partial t)(\omega, \tau)| \leq g(\omega) \qquad \text{for all } \omega \in \Omega.$$

(c) Take $t_0 \in (a,b)$ and let $(t_n)_{n\geq 1}$ be a sequence in (a,b) such that $t_n \to t_0$ and define

$$h(\omega, t) := \frac{f(\omega, t) - f(\omega, t_0)}{t - t_0}.$$

Then by (b) and the conditions in this exercise, h satisfies all the conditions listed in Problem 8.2.

Next, we note that by linearity of the integral we have that

$$\frac{F(t_n) - F(t_0)}{t_n - t_0} = \int_\Omega \frac{f(\omega, t_n) - f(\omega, t_0)}{t_n - t_0} \,\mu(d\omega) = \int_\Omega h(\omega, t_n)\,\mu(\mathrm{d}\omega).$$

Now by Problem 8.2 it holds that

$$\lim_{t_n \to t_0} \int_\Omega h(\omega, t_n)\,\mu(\mathrm{d}\omega) = \int_\Omega \lim_{t_n\to t_0} h(\omega, t_n)\,\mu(\mathrm{d}\omega) = \int_\Omega \frac{\partial f}{\partial t}(\omega, t_0)\,\mu(\mathrm{d}\omega).$$

Since $t_0 \in (a,b)$ and the sequence $(t_n)_{n\geq 1}$ were arbitrary, we conclude that F is indeed differentiable on (a,b) with

$$\frac{\partial F}{\partial t}(t_0) := \lim_{t_n \to t_0} \frac{F(t_n) - F(t_0)}{t_n - t_0} = \int_\Omega \frac{\partial f}{\partial t}(\omega, t_0)\,\mu(\mathrm{d}\omega).$$

Problem 8.4

(a) Note that the integrand $f_n(x) = \frac{1+nx^2}{(1+x^2)^n}$ is continuous on $[0,1]$ and non-negative. Hence, the Riemann integral and Lebesgue integral coincide, i.e.,

$$\int_0^1 f_n(x)\,\mathrm{d}x = \int_{[0,1]} f_n\,\mathrm{d}\lambda.$$

Observe that we have the following pointwise limit

$$\lim_{n\to\infty} f_n(x) = \begin{cases} 1 & \text{if } x = 0, \\ 0 & \text{if } x \in (0,1], \end{cases}$$

i.e., $\lim_{n\to\infty} f_n = 0$ λ-almost everywhere. Moreover, $f_n(x) \le 1$ for every $x \in [0,1]$ and $n \ge 1$. Since the constant function $g \equiv 1$ is λ-integrable on $[0,1]$, it is a valid dominator. Hence, the DCT gives

$$\lim_{n\to\infty} \int_0^1 f_n(x)\,\mathrm{d}x = \lim_{n\to\infty} \int_{[0,1]} f_n\,\mathrm{d}\lambda = \int_{[0,1]} \lim_{n\to\infty} f_n\,\mathrm{d}\lambda = 0$$

(b) For the purpose of convergence, we consider $n \ge 3$. Note that the integrand $f_n(x) = \frac{x^{n-2}}{1+x^n}\cos\left(\frac{\pi x}{n}\right)$ is continuous on $(0,+\infty)$ with pointwise limit

$$\lim_{n\to\infty} f_n(x) = \begin{cases} 0 & \text{if } x \in (0,1), \\ 1/2 & \text{if } x = 1, \\ 1/x^2 & \text{if } x > 1, \end{cases}$$

Setting the function

$$g(x) = \begin{cases} 1 & \text{for } x \in (0,1), \\ \frac{1}{x^2} & \text{for } x \ge 1, \end{cases}$$

we see that $f_n \le g$ λ-almost everywhere in $(0,+\infty)$ and for all $n \ge 3$. Indeed, for $x \ge 1$, we obtain

$$|f_n(x)| \le \left|\frac{x^{n-2}}{1+x^n}\cos\left(\frac{\pi x}{n}\right)\right| \le \frac{x^{n-2}}{1+x^n} \le \frac{x^{n-2}}{x^n} = \frac{1}{x^2},$$

while for $x \in (0,1)$, we have

$$|f_n(x)| \le \left|\frac{x^{n-2}}{1+x^n}\cos\left(\frac{\pi x}{n}\right)\right| \le \frac{x^{n-2}}{1+x^n} \le 1.$$

Notice that g is non-negative and λ-integrable on $(0,+\infty)$. Indeed, using the MCT,

$$\begin{aligned}\int_{(0,+\infty)} g\,\mathrm{d}\lambda &= \int_{(0,1)} g\,\mathrm{d}\lambda + \int_{(1,+\infty)} g\,\mathrm{d}\lambda = 1 + \lim_{n\to\infty} \int_{(1,n)} g\,\mathrm{d}\lambda \\ &= 1 + \lim_{n\to\infty} \int_1^n \frac{1}{x^2}\,\mathrm{d}x = 1 + \lim_{n\to\infty}\left(1 - \frac{1}{n}\right) = 2 < +\infty.\end{aligned}$$

To conclude, we apply DCT to deduce that the limit is 1.

Problem 8.5

Problem 8.6

(a)

(b) By definition
$$\lim_{n\to\infty}\left|\int_{\mathbb{R}} f\,\mathrm{d}\mu_n - \int_{\mathbb{R}} f\,\mathrm{d}\mu\right| \leq \varepsilon,$$
implies that for any $\delta > 0$
$$\left|\int_{\mathbb{R}} f\,\mathrm{d}\mu_n - \int_{\mathbb{R}} f\,\mathrm{d}\mu\right| < \varepsilon + \delta,$$
holds for large enough n. Note that this holds for any $\varepsilon, \delta > 0$.

Now pick $\eta > 0$ and set $\varepsilon = \eta/2 = \delta$, then the above inequality implies that
$$\lim_{n\to\infty}\left|\int_{\mathbb{R}} f\,\mathrm{d}\mu_n - \int_{\mathbb{R}} f\,\mathrm{d}\mu\right| = 0.$$

(c) Consider the sequence of sets $A_n = \mathbb{R} \setminus [-n, n]$. Then $A_n \supset A_{n+1}$ and $A_n \downarrow \emptyset$. Hence, it follows from Proposition 2.12 2) that $\lim_{n\to\infty} \mu(A_n) = 0$. Thus, there exists a N such that $\mu(A_n) < \varepsilon/(2M)$ holds for all $n \geq N$. We can then take any $\alpha > N$.

(d) The function
$$g(x) = \mathbb{1}_{[-\alpha,\alpha]}(x) + \mathbb{1}_{(-(\alpha+1),-\alpha)}(x)\,(x + (\alpha + 1)) + \mathbb{1}_{(\alpha,\alpha+1)}(x)\,(-x + \alpha + 1)$$
does the trick. This is simply a linear increase from zero to one from $-(\alpha + 1)$ to $-\alpha$ and from $\alpha + 1$ to α.

(e) Observe that g is a non-negative continuous bounded function that is zero outside the interval $[-(\alpha + 1), \alpha + 1]$, and thus we can apply (3). Using linearity of the integral, the fact that $|f| \leq M$ and the definition of g, we get
$$\begin{aligned}\left|\int_{\mathbb{R}} f\,\mathrm{d}\mu - \int_{\mathbb{R}} fg\,\mathrm{d}\mu\right| = \left|\int_{\mathbb{R}} f(1-g)\,\mathrm{d}\mu\right| &\leq M\int_{\mathbb{R}}(1-g)\,\mathrm{d}\mu \\ &\leq M\int_{\mathbb{R}}(1-g)\,\mathrm{d}\mu \\ &= M\left(1 - \int_{\mathbb{R}} g\,\mathrm{d}\mu\right) \\ &\leq M\mu(\mathbb{R}\backslash[-\alpha,\alpha]) < \frac{\varepsilon}{2}.\end{aligned}$$

(f) Again, using linearity of the integral and the fact that $|f| \leq M$ we get

$$\begin{aligned}\left|\int_{\mathbb{R}} f\,\mathrm{d}\mu_n - \int_{\mathbb{R}} fg\,\mathrm{d}\mu_n\right| &= \left|\int_{\mathbb{R}} f(1-g)\,\mathrm{d}\mu_n\right| \leq M\int_{\mathbb{R}}(1-g)\,\mathrm{d}\mu_n \\ &\leq M\int_{\mathbb{R}}(1-g)\,\mathrm{d}\mu_n = M\left(1-\int_{\mathbb{R}} g\,\mathrm{d}\mu_n\right)\end{aligned}$$

Now observe that the integral in the last term converges to $\int_{\mathbb{R}} g\,\mathrm{d}\mu$ by (3). Thus, we obtain

$$\limsup_{n\to\infty}\left|\int_{\mathbb{R}} f\,\mathrm{d}\mu_n - \int_{\mathbb{R}} fg\,\mathrm{d}\mu_n\right| \leq M\int_{\mathbb{R}}(1-g)\,\mathrm{d}\mu \leq M\mu(\mathbb{R}\backslash[-\alpha,\alpha]) < \frac{\varepsilon}{2}.$$

(g) Recall that

$$\begin{aligned}\left|\int_{\mathbb{R}} f\,\mathrm{d}\mu_n - \int_{\mathbb{R}} f\,\mathrm{d}\mu\right| \leq \left|\int_{\mathbb{R}} f\,\mathrm{d}\mu_n - \int_{\mathbb{R}} fg\,\mathrm{d}\mu_n\right| &+ \left|\int_{\mathbb{R}} f\,\mathrm{d}\mu - \int_{\mathbb{R}} fg\,\mathrm{d}\mu\right| \\ &+ \left|\int_{\mathbb{R}} fg\,\mathrm{d}\mu_n - \int_{\mathbb{R}} fg\,\mathrm{d}\mu\right|.\end{aligned}$$

For the first two terms, the (e) and (f) imply that the $\limsup_{n\to\infty}$ is bounded by $\varepsilon/2$. For the third term we not that fg is a continuous bounded function and hence this term converges to zero by our assumption that (3) holds.

Together we then have that

$$\limsup_{n\to\infty}\left|\int_{\mathbb{R}} f\,\mathrm{d}\mu_n - \int_{\mathbb{R}} f\,\mathrm{d}\mu\right| < \varepsilon,$$

which implies the result.

9 Convergence of random variables

Problem 9.2

(a) For this let $h_t(x) = \mathbf{1}_{(-\infty,t]}$ and note that

$$F_n(t) = (X_n)_\# \mathbb{P}_n((-\infty, x]) = \int_{\mathbb{R}} \mathbb{1}_{(-\infty,t]} \, \mathrm{d}(X_n)_\# \mathbb{P}_n = \int_{\mathbb{R}} h_t \, \mathrm{d}\mu_n.$$

and similarly $F(t) = \int_{\mathbb{R}} h_t \, \mathrm{d}\mu$

(b) The function h is discontinuous only at t, i.e. $\mathcal{C}_h = \mathbb{R} \setminus \{t\}$.

Moreover, for any $\varepsilon > 0$

$$\mu((t-\varepsilon, t+\varepsilon)) = \mu((t-\varepsilon, t]) + \mu((t, t+\varepsilon)) = F(t) - F(t-\varepsilon) + F(t+\varepsilon) - F(t).$$

Since F is continuous at t, the right hand side goes to zero as $\varepsilon \to 0$. Therefore

$$\mu(\{t\}) = \lim_{\varepsilon \to 0} \mu((t-\varepsilon, t+\varepsilon)) = 0,$$

which implies that $\mu(\mathcal{C}_h) = 1$.

(c) The result follows by applying condition (2) in Theorem 8.6.

(d) Without loss of generality assume $x \in I_\ell = (a_\ell, b_\ell]$ for some $1 \leq \ell \leq L$. Then it holds that $\hat{g}(x) = g(b_\ell)$.

Moreover, since $|x - b_\ell| < \delta$ we have that

$$|g(x) - \hat{g}(x)| = |g(x) - g(b_\ell)| < \varepsilon.$$

(e) Let $M = L$, $\beta_\ell = \sum_{t=1}^{\ell} h(b_t)$ and $t_\ell = b_\ell$. Then

$$\hat{g} := \sum_{\ell=1}^{L} \beta_\ell \mathbf{1}_{(-\infty, b_\ell]}.$$

(f) Using the representation in (e) we get

$$\begin{aligned}\lim_{n\to\infty}\mathbb{E}[\hat{g}(X_n)] &= \lim_{n\to\infty}\sum_{\ell=1}^{L}\beta_\ell\int_{\mathbb{R}}\mathbf{1}_{X_n^{-1}((-\infty,b_\ell])}\,\mathrm{d}\mathbb{P}\\ &= \lim_{n\to\infty}\sum_{\ell=1}^{L}\beta_\ell(X_n)_{\#}\mathbb{P}((-\infty,b_\ell])\\ &= \lim_{n\to\infty}\sum_{\ell=1}^{L}\beta_\ell F_n(b_\ell)\\ &= \sum_{\ell=1}^{L}F(b_\ell)=\mathbb{E}[\hat{g}(X)].\end{aligned}$$

(g) First we write

$$\begin{aligned}\|\mathbb{E}[g(X_n)]-\mathbb{E}[g(X)]\| \leq \|\mathbb{E}[g(X_n)]-\mathbb{E}[\hat{g}(X_n)]\| + \|\mathbb{E}[g(X)]-\mathbb{E}[\hat{g}(X)]\|\\ + \|\mathbb{E}[\hat{g}(X_n)]-\mathbb{E}[\hat{g}(X)]\|.\end{aligned}$$

We have shown in (f) that the last term goes to zero as $n\to\infty$.

Next, using (d) it follows that the other two terms are bounded by ε.

Since ε was arbitrary we conclude that

$$\lim_{n\to\infty}\mathbb{E}[g(X_n)]=\mathbb{E}[g(X)].$$

(h) This now follows from Theorem 8.6 (3).

Problem 9.3

The main idea is to use the equivalent version of convergence in distribution.
Suppose that $X_n\xrightarrow{\mathbb{P}}X$ and define $Y_n=|X_n-X|$. We need to show that $\mathbb{P}(Y_n>\varepsilon)\to 0$ holds for any $\varepsilon>0$. First recall that $X_n\xrightarrow{\mathbb{P}}X$ is defined as weak convergence of Y_n to the constant zero random variable. By Lemma 8.2 this is equivalent to

$$\lim_{n\to\infty}\mathbb{P}(Y_n\leq t)=\mathbb{P}(0\leq t),$$

for all continuity points of the function $\omega\mapsto 0$. We now note that any $\varepsilon>0$ is a continuity point of this function. Hence, we get

$$\lim_{n\to\infty}\mathbb{P}(Y_n>\varepsilon)=1-\lim_{n\to\infty}\mathbb{P}(Y_n\leq\varepsilon)=1-\mathbb{P}(0\leq\varepsilon)=0$$

Now we prove the other implication. So suppose that $\mathbb{P}(Y_n>\varepsilon)\to 0$ holds for any $\varepsilon>0$. We then have to prove that $(Y_n)_{\#}\mathbb{P}\Rightarrow 0_{\#}\mathbb{P}$. Due to Lemma 8.2 it is enough to show that

$$\lim_{n\to\infty}\mathbb{P}(Y_n\leq t)=\mathbb{P}(0\leq t)=\mathbb{1}_{t\geq 0},$$

holds for all continuity points t of the function $\omega \mapsto 0$. Notice that the only non-continuity point is 0. Moreover, for all $t < 0$ we have that $\mathbb{P}(Y_n \leq t) = 0$ since $Y_n \geq 0$ almost every-where. Finally, for all $t > 0$ we have

$$\lim_{n\to\infty} \mathbb{P}(Y_n \leq t) = 1 - \lim_{n\to\infty} \mathbb{P}(Y_n > t) = 1 = \mathbb{P}(0 \leq t).$$

Problem 9.5 Suppose that $X_n \overset{\text{a.s.}}{\to} X$. Then by Lemma 5.2.16 this is equivalent to $\mathbb{P}(\|X_n - X\| > \varepsilon \text{ i.o.}) = 0$ for all $\varepsilon > 0$.
For now fix an $\varepsilon > 0$ and write $A_n := \{\|X_n - X\| > \varepsilon\}$. Recall that

$$\{A_n \text{ i.o.}\} = \bigcap_{k=1}^{\infty} \bigcup_{k\geq n} A_n$$

and note two things:

1. The sets $B_k := \bigcup_{n\geq k} A_n$ are non-increasing, i.e. $B_k \supset B_{k+1}$, and
2. $\mathbb{P}(A_k) \leq \mathbb{P}(\bigcup_{n\geq k} A_n) = \mathbb{P}(B_k)$.

We then have that:

$$\begin{aligned}
0 &= \mathbb{P}(\{A_n \text{ i.o.}\}) && \text{by assumption} \\
&= \mathbb{P}(\bigcap_{k=1}^{\infty} B_k) && \text{by Lemma 5.2.16} \\
&= \lim_{k\to\infty} \mathbb{P}(B_k) && \text{by continuity form above (Proposition 2.2.15)} \\
&\geq \lim_{k\to\infty} \mathbb{P}(A_k) && \text{by (b).}
\end{aligned}$$

Problem 9.6

(a) Define the sets

$$B_j := \bigcup_{i\geq j} A_i, \qquad j \in \mathbb{N}.$$

Clearly the sequence $(B_j)_{j\in\mathbb{N}}$ is decreasing and $\{A_n \text{ i.o.}\} \subset B_j$ for every $j \in \mathbb{N}$.

By assumption, and the σ-subadditivity of $\mathbb{P}$,

$$\mathbb{P}(B_1) = \mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mathbb{P}(A_i) < +\infty.$$

Moreover, the summability also gives

$$\lim_{j\to\infty} \mathbb{P}(B_j) \leq \limsup_{j\to\infty} \sum_{i=j}^{\infty} \mathbb{P}(A_i) = 0.$$

Hence, by the continuity from above of μ, we obtain

$$\mathbb{P}(\{A_n \text{ i.o.}\}) \leq \mathbb{P}\left(\bigcup_{j=1}^{\infty} B_j\right) = \lim_{j\to\infty} \mathbb{P}(B_j) = 0,$$

i.e., $\{A_n \text{ i.o.}\}$ is a null set. In other words, $\mathbb{P}$-almost every ω is in only finitely many A_n.

(b) We will prove that

$$\mathbb{P}(\Omega \setminus \{A_n \text{ i.o.}\}) = 0,$$

from which the result follows since $\mathbb{P}(\Omega) = 1$.

First note that

$$\Omega \setminus \{A_n \text{ i.o.}\} = \bigcup_{k\geq 1}\left(\bigcup_{n\geq k} A_n\right)^c = \bigcup_{k\geq 1}\bigcap_{n\geq k} A_n^c.$$

Next, since A_n are mutually exclusive, so are A_n^c. Thus, for any $k \geq 1$ we have that

$$\begin{aligned}\mathbb{P}\left(\bigcap_{n\geq k} A_n^c\right) &= \prod_{n\geq k} \mathbb{P}(A_n^c) = \prod_{n\geq k}(1 - \mathbb{P}(A_m)\\ &\leq \prod_{n\geq k} e^{-\mathbb{P}(A_n)} = e^{-\sum_{n\geq k}\mathbb{P}(A_n)} = 0.\end{aligned}$$

Here we used that for any $0 \leq x \leq 1$ it holds that $1 - x \leq e^{-x}$.

Finally, using σ-subadditivity we conclude that

$$\mathbb{P}(\Omega \setminus \{A_n \text{ i.o.}\}) = \mathbb{P}\left(\bigcup_{k\geq 1}\bigcap_{n\geq k} A_n^c\right) \leq \sum_{k\geq 1}\mathbb{P}\left(\bigcap_{n\geq k} A_n^c\right) = 0.$$

Problem 9.7

Fix $\varepsilon > 0$ and define $A_n(\varepsilon) : \{|X_n - X| > \varepsilon\}$. Then the assumption translates to

$$\sum_{n\geq 1}\mathbb{P}(A_n(\varepsilon)) < \infty.$$

By Lemma 8.11 1) this then implies that $\mathbb{P}(A_n(\varepsilon) \text{ i.o.}) = 0$. Since $\varepsilon > 0$ was arbitrary, Lemma 8.9 now implies that $X_n \stackrel{\text{a.s.}}{\to} X$.

10 L^p-spaces

Problem 10.3 (5 pts)
Let $(\Omega, \mathcal{F}, \mu)$ be a finite measure space.

(a) (2 pts) Let $p \leq q$, $p, q \in [1, \infty)$. We will show that $\Phi_p(f) \leq \Phi_q(f)$ whenever $\Phi_q(f)$ is finite, otherwise, there is nothing to show. Using Jensen's inequality, we immediately find

$$\int_\Omega |f|^q \, \frac{d\mu}{\mu(\Omega)} = \int_\Omega \left(|f|^p\right)^{\frac{q}{p}} \frac{d\mu}{\mu(\Omega)} \geq \left(\int_\Omega |f|^p \, \frac{d\mu}{\mu(\Omega)}\right)^{\frac{q}{p}}.$$

Taking the q-th root then yields the claim.

(b) (3 pt) If $f \in L^\infty(\mu)$, then clearly $\Phi_p(f) \leq \|f\|_\infty$. We are now left to show that the upper bound is saturated. W.l.o.g. we assume that $\|f\|_\infty = 1$, otherwise, we can take $f/\|f\|_\infty$, since

$$\frac{1}{\|f\|_\infty} \Phi_p(f) = \Phi_p\left(\frac{f}{\|f\|_\infty}\right) \qquad \text{for every } p \in [1, \infty).$$

In the case $\|f\|_\infty = 1$, we have find that

$$0 \leq a_p := \int_\Omega |f|^p \, \frac{d\mu}{\mu(\Omega)} \leq 1 \qquad \text{independently of } p.$$

Now, since a_p is increasing due to (a) and bounded from above, we deduce that a_p has a limit (Analysis 1). By the squeeze theorem, we then conclude that $\lim_{p\to\infty} (a_p)^{1/p} = 1$.

11 The Radon-Nikodym derivative and conditional expectation and probability

Problem 11.1 (5 pts)
Let $\nu \ll \mu$. We show the required statement by contradiction. Therefore, let us suppose otherwise, i.e., there exists $\varepsilon > 0$ such that for any $\delta > 0$, there is some $A \in \mathcal{F}$ for which

$$\mu(A) < \delta, \quad \text{but} \ \ \nu(A) \geq \varepsilon. \qquad \text{(1 pts)}$$

We can choose a sequence of decreasing sets A_n, i.e., $A_{n+1} \subset A_n \subset A_1$, such that $\mu(A_n) < 1/n$ and $\nu(A_n) \geq \varepsilon$ for all $n \geq 1$ (1 pts). By continuity from above, we have that

$$\mu\Big(\bigcap_{n\geq 1} A_n\Big) = \lim_{n\to\infty} \mu(A_n) = 0, \qquad \nu\Big(\bigcap_{n\geq 1} A_n\Big) = \lim_{n\to\infty} \nu(A_n) \geq \varepsilon \qquad \text{(1 pts)}$$

i.e., $\bigcap_{n\geq 1} A_n$ is a μ-null set (1 pts). Since ν is absolutely continuous w.r.t. μ, then $\bigcap_{n\geq 1} A_n$ is also a ν-null set, which contradicts with the deduced lower bound. (1 pts)

Problem 11.2 (3 pts)
Suppose that $f, g : \Omega \to \mathbb{R}$ are $\mathcal{H}$-measurable functions satisfying

$$\int_B f \, d\mathbb{P} = \int_B g \, d\mathbb{P} \qquad \text{for all } B \in \mathcal{H}.$$

Since f and g are $\mathcal{H}$-measurable, so are the sets $\{f > g\}$ and $\{f < g\}$ (1 pts). Then,

$$\int_{\{f>g\}} |f - g| \, d\mathbb{P} = \int_{\{f>g\}} (f - g) \, d\mathbb{P} = 0. \qquad \text{(0.5 pts)}$$

Similarly, we have that

$$\int_{\{f<g\}} |f - g| \, d\mathbb{P} = \int_{\{f<g\}} (g - f) \, d\mathbb{P} = 0. \qquad \text{(0.5 pts)}$$

Altogether, we find

$$\int_\Omega |f - g| \, d\mathbb{P} = 0 \quad \Longrightarrow \quad f = g \ \ \mathbb{P}\text{-almost everywhere.} \qquad \text{(1 pts)}$$

Problem 11.3

(a) By definition we have that

$$\int_B \mathbb{E}[X|\mathcal{H}]\,\mathrm{d}\mathbb{P} = \int_B X\,\mathrm{d}\mathbb{P},$$

holds for all $B \in \mathcal{H}$. Since by assumption both $\mathbb{E}[X|\mathcal{H}]$ and X are $\mathcal{H}$-measurable, the result follows from problem 8.2.

(b) Note that $a\mathbb{E}[X|\mathcal{H}]$ is $\mathcal{H}$-measurable. Moreover,

$$\int_B a\mathbb{E}[X|\mathcal{H}]\,\mathrm{d}\mathbb{P} = a\int_B \mathbb{E}[X|\mathcal{H}]\,\mathrm{d}\mathbb{P} = a\int_B X\,\mathrm{d}\mathbb{P} = \int_B aX\,\mathrm{d}\mathbb{P}.$$

This proves the claim.

(c) Similarly to the previous point, we first note that since $\mathbb{E}[X|\mathcal{H}]$ and $\mathbb{E}[Y|\mathcal{H}]$ are $\mathcal{H}$-measurable so is $\mathbb{E}[X|\mathcal{H}] + \mathbb{E}[Y|\mathcal{H}]$. The result then follows because

$$\begin{aligned}\int_B \mathbb{E}[X|\mathcal{H}] + \mathbb{E}[Y|\mathcal{H}]\,\mathrm{d}\mathbb{P} &= \int_B \mathbb{E}[X|\mathcal{H}]\,\mathrm{d}\mathbb{P} + \int_B \mathbb{E}[Y|\mathcal{H}]\,\mathrm{d}\mathbb{P}\\ &= \int_B X\,\mathrm{d}\mathbb{P} + \int_B Y\,\mathrm{d}\mathbb{P} = \int_B X + Y\,\mathrm{d}\mathbb{P}.\end{aligned}$$

(d) First we observe that for any $B \in \mathcal{H}$

$$\int_B \mathbb{E}[X|\mathcal{H}]\,\mathrm{d}\mathbb{P} = \int_B X\,\mathrm{d}\mathbb{P} \leq \int_B Y\,\mathrm{d}\mathbb{P} = \int_B \mathbb{E}[Y|\mathcal{H}]\,\mathrm{d}\mathbb{P}.$$

Now consider the event $A := \{\mathbb{E}[X|\mathcal{H}] > \mathbb{E}[Y|\mathcal{H}]\} \in \mathcal{H}$. If this event has non-zero measure then it would follow that

$$\int_A \mathbb{E}[X|\mathcal{H}]\,\mathrm{d}\mathbb{P} > \int_A \mathbb{E}[Y|\mathcal{H}]\,\mathrm{d}\mathbb{P},$$

which is a contradiction. Hence we conclude that $\mathbb{E}[X|\mathcal{H}] \leq \mathbb{E}[Y|\mathcal{H}]$ holds $\mathbb{P}$-almost everywhere.

Problem 11.4

(a) This follows by repeating the step for the solution to Problem 4.8 a).

(b) We start by observing that the result will directly follow from Theorem 2.15 if we can show that the σ-algebra $\sigma(X)$ satisfies the two properties.

The first one is immediate, from the fact that $X^{-1}(A) \cap X^{-1}(B) = X^{-1}(A\ capB)$. For the second one consider the intervals $I_n = (-n, n)$ and define $A_n := X^{-1}(I_n) \in \sigma(X)$. Since $\bigcup_{n\in\mathbb{N}} I_n = \mathbb{R}$ it follows that $\bigcup_{n\in\mathbb{N}} A_n = \Omega$. Moreover since $|X| \leq n$ on the set A_n it holds that

$$\mu(A_n) = \nu_X(A_n) = \int_{A_n} X\,\mathrm{d}\mathbb{P} \leq n\int_{A_n} \mathrm{d}\mathbb{P} = n\mathbb{P}(A_n) \leq n < \infty.$$

Thus the second condition of Theorem 2.15 is also satisfied and the result now follows.

Problem 11.5

The solution to this problem will closely follow the proof of Lemma 11.7. To this end let $g(x, y)$ denote the conditional density of X given $Y = y$ and define the function $\phi(y) := \int_{\mathbb{R}} h(x)g(x, y)\,\lambda(\mathrm{d}x)$. We now need to show that for any $A \in \mathcal{B}_{\mathbb{R}}$ it holds that

$$\int_{Y^{-1}(A)} \phi(Y)\,\mathrm{d}\mathbb{P} = \mathbb{E}[\mathbb{1}_A(Y)h(X)].$$

Using the change of variables formula and the definition of ϕ and g we get

$$\begin{aligned}
\int_{Y^{-1}(A)} \phi(Y)\,\mathrm{d}\mathbb{P} &= \int_{\Omega} \mathbb{1}_{Y^{-1}(A)}\phi(Y)\,\mathrm{d}\mathbb{P} \\
&= \int_{\mathbb{R}} \int_{A} (y)\phi(y) f_Y(y)\,\lambda(\mathrm{d}y) \\
&= \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbb{1}_A(y)h(x)g(x, y)f_Y(y)\,\lambda(\mathrm{d}x)\,\lambda(\mathrm{d}y) \\
&= \int_{\mathbb{R}^2} \mathbb{1}_A(y)h(x)f(x, y)\,\lambda(\mathrm{d}x)\,\lambda(\mathrm{d}y) \\
&= \mathbb{E}[\mathbb{1}_A(Y)h(X)].
\end{aligned}$$

Problem 11.6

(a) First note that for any $\ell \geq 0$

$$\mathbb{P}(X > \ell) = \int_{\ell} \nu e^{-\nu x}\,\lambda(dx) = e^{-\nu \ell}.$$

By Lemma 11.5 we have that

$$\mathbb{P}(X > t+s | X > t) = \frac{\mathbb{P}(X > t + s, X > t)}{\mathbb{P}(X > t)} = \frac{\mathbb{P}(X > t + s)}{\mathbb{P}(X > t)} = \frac{e^{-\nu(t+s)}}{e^{-\nu t}} = e^{-\nu s} = \mathbb{P}(X > s).$$

(b) Define the function $f(x, y) = \mathbb{1}_{x+y \leq t}$. By Lemma 11.8 we then have that

$$\mathbb{P}(X + Y \leq t | Y = y) = \mathbb{E}[f(X, Y) | Y = y] = \mathbb{E}[f(X, y)].$$

Using the probability density function we then get

$$\mathbb{E}[f(X, y)] = \int_{\mathbb{R}} f(x, y)\rho(x)\,\lambda(dx) = \mathbb{1}_{y<t} \int_0^{t-y} \nu e^{-\nu x}\,\lambda(dx) = \mathbb{1}_{y<t}(1 - e^{-\nu(t-y)}).$$

(c) We have that

$$\begin{aligned}\mathbb{P}(X+Y\le t) &= \mathbb{E}[\mathbb{P}(X+Y\le t|Y)]\\ &= \int_{\mathbb{R}} \rho(y)\mathbb{P}(X+Y\le t|Y=y)\,\lambda(dx)\\ &= \int_{\mathbb{R}} \mathbb{1}_{0\le y<t}(1-e^{-\nu(t-y)})\nu e^{-\nu y}\,\lambda(dy)\\ &= \int_0^t \nu e^{-\nu y} - e^{-\nu t}\,\lambda(dy)\\ &= 1-e^{-\nu t}-te^{-\nu t}.\end{aligned}$$

Problem 11.7

(a) (3 pts) **2 pts** We first observe that

$$\int_B Z\,\mathrm{d}\mathbb{P} = \int_\Omega f\circ[(X,Y)\circ\Delta\mathbb{1}_B]\,\mathrm{d}\mathbb{P} = \int_{\mathbb{R}^2} f\,\mathrm{d}[(X,Y)\circ\Delta\mathbb{1}_B]_\#\mathbb{P}.$$

On the other hand

$$\int_{\Omega\times B} f\circ(X,Y)\,\mathrm{d}\mathbb{P}\otimes\mathbb{P} = \int_{\Omega\times\Omega} f\circ(X,Y)\mathbb{1}_{\Omega\times B}\,\mathrm{d}\mathbb{P}\otimes\mathbb{P} = \int_{\mathbb{R}^2} f\,\mathrm{d}[(X,Y)\mathbb{1}_{\Omega\times B}]_\#\mathbb{P}\otimes\mathbb{P}.$$

Thus, the result follows if we can show that

$$[(X,Y)\mathbb{1}_{\Omega\times B}]_\#\mathbb{P}\otimes\mathbb{P} = [(X,Y)\circ\Delta\mathbb{1}_B]_\#\mathbb{P}$$

as measures on $\mathcal{B}^2=\mathcal{B}\otimes\mathcal{B}$.

1 pt Since both measure are finite it follows from Theorem 2.15 that is suffices to check equality on the generator set $\mathcal{B}\times\mathcal{B}$.

2 pts To this end, let $C, D\in\mathcal{B}$. Then it holds that

$$\begin{aligned}[(X,Y)\circ\Delta\mathbb{1}_B]^{-1}(C\times D) &= \{\omega\in\Omega\,:\,\omega\in B, X(\omega)\in C, Y(\omega)\in D\}\\ &= X^{-1}(C)\cap Y^{-1}(D)\cap B.\end{aligned}$$

Recall that $B\in\sigma(Y)$ so that also $Y^{-1}(D)\cap B\in\sigma(Y)$. Thus, since X and Y are independent

$$\begin{aligned}[(X,Y)\circ\Delta\mathbb{1}_B]_\#\mathbb{P}(C\times D) &= \mathbb{P}(X^{-1}(C)\cap Y^{-1}(D)\cap B)\\ &= \mathbb{P}(X^{-1}(C))\mathbb{P}(Y^{-1}(D)\cap B).\end{aligned}$$

2 pts On the other hand, since

$$[(X,Y)\mathbb{1}_{\Omega\times B}]^{-1}(C\times D) = X^{-1}(C)\times Y^{-1}(D)\cap B,$$

we have that

$$
\begin{aligned}
[(X,Y)\mathbb{1}_{\Omega\times B}]_{\#}\mathbb{P}\otimes\mathbb{P}(C\times D) &= \mathbb{P}\otimes\mathbb{P}(X^{-1}(C)\times Y^{-1}(D)\cap B) \\
&= \mathbb{P}(X^{-1}(C))\mathbb{P}(Y^{-1}(D)\cap B).
\end{aligned}
$$

We therefore conclude that both measures are equal on the generator set of $\mathcal{B}^2$ and hence are equal on the entire σ-algebra.

(b) (3 pts) **1 pt** As noted in the proof of Lemma 11.8 the measures $(X,Y)_{\#}\mathbb{P}\otimes\mathbb{P}$ and $X_{\#}\mathbb{P}\otimes Y_{\#}\mathbb{P}$ agree on the generator set of $\mathcal{B}^2$. Thus, using Theorem 11.5 the are equal on the entire σ-algebra.

2 pts Using Fubini-Tonelli twice we then have that

$$
\begin{aligned}
\int_{R\times C} f\mathrm{d}(X,Y)_{\#}\mathbb{P}\otimes\mathbb{P} &= \int_{R\times C} f\,\mathrm{d}X_{\#}\mathbb{P}\otimes Y_{\#}\mathbb{P} \\
&= \int_{\mathbb{R}}\int_{C} f(x,y)Y_{\#}\mathbb{P}(\mathrm{d}y)X_{\#}\mathbb{P}(\mathrm{d}x) \\
&= \int_{C}\int_{\mathbb{R}} f(x,y)X_{\#}\mathbb{P}(\mathrm{d}X)Y_{\#}\mathbb{P}(\mathrm{d}y).
\end{aligned}
$$